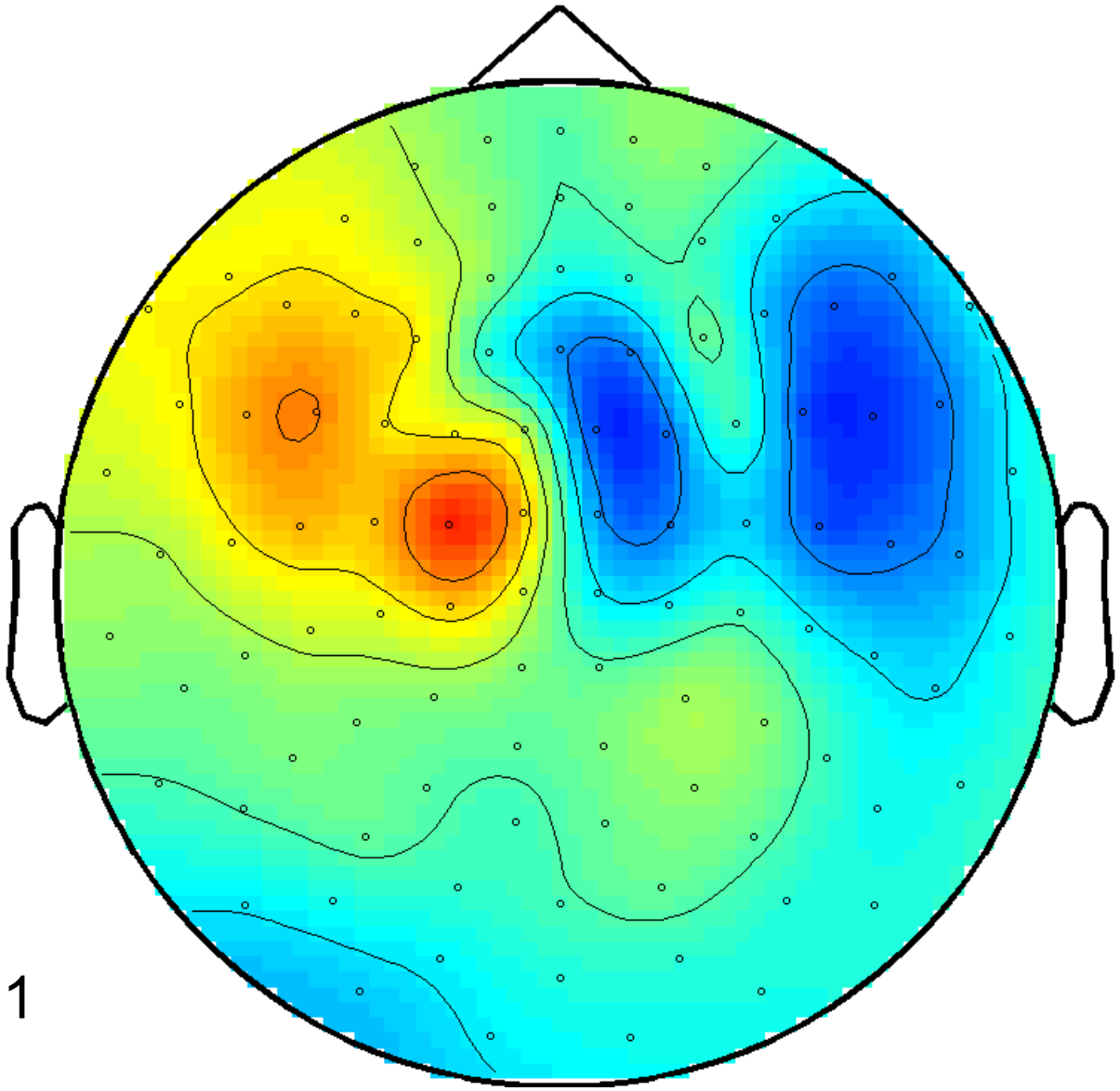
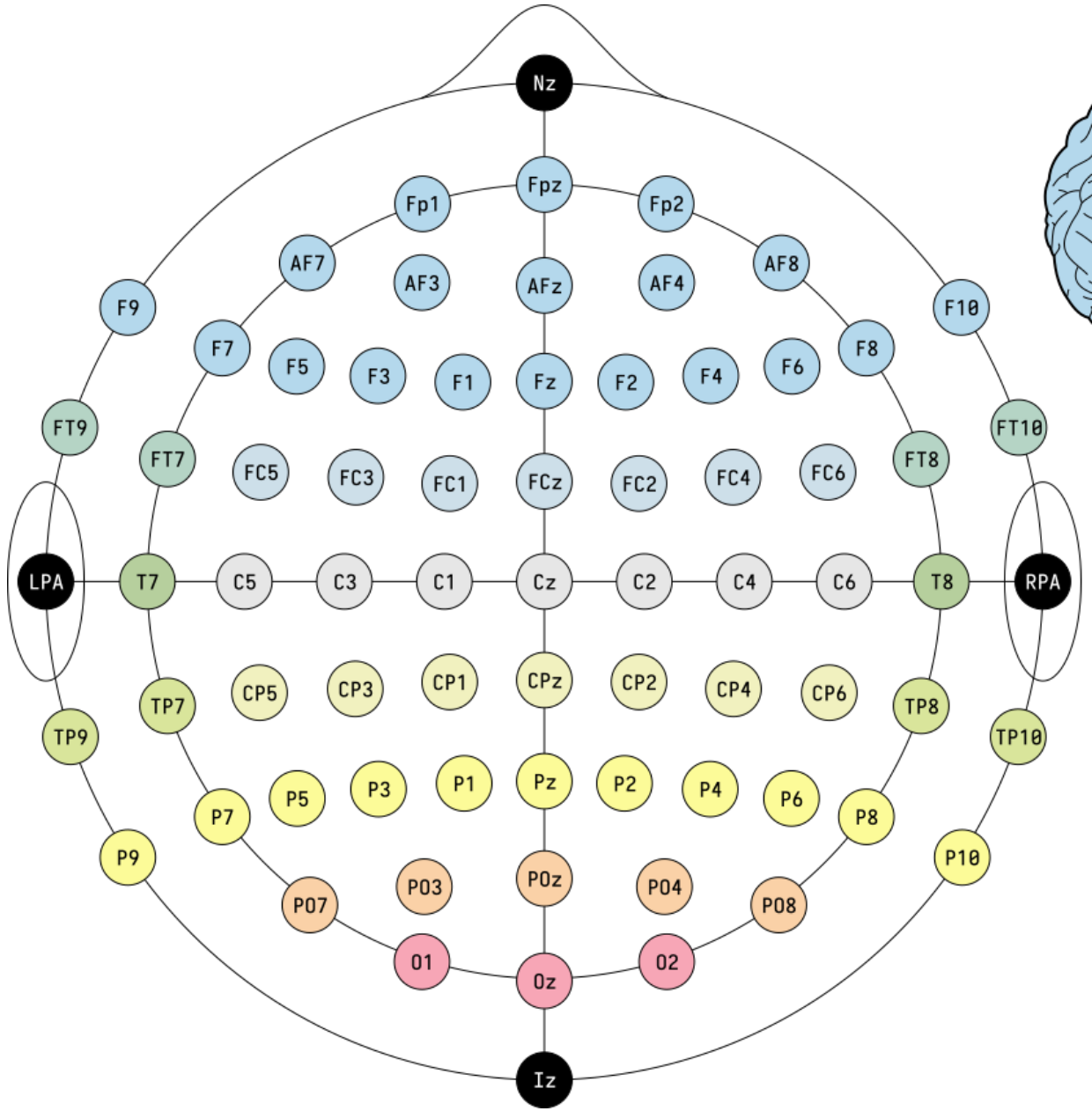


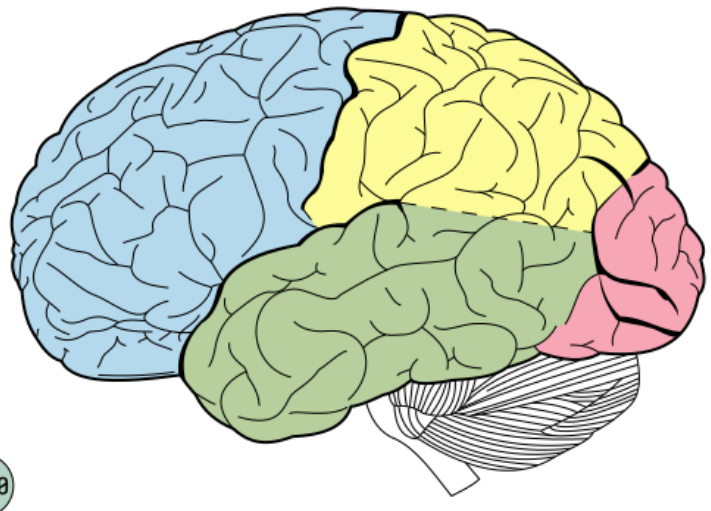
Deep learning MO74481
10/03/2026

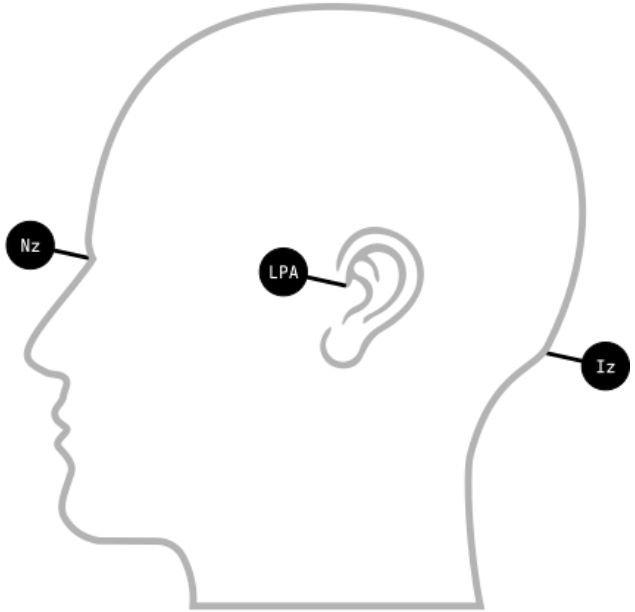
UAB
Universitat Autònoma
de Barcelona

Project 2

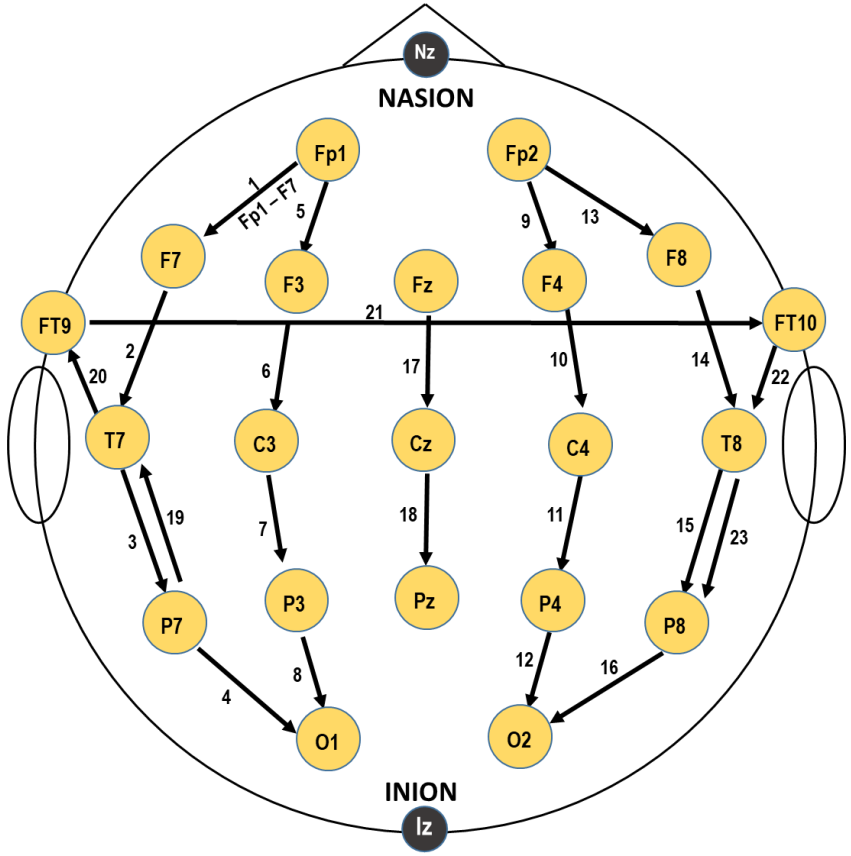
Detection of epileptic seizures in EEG signals



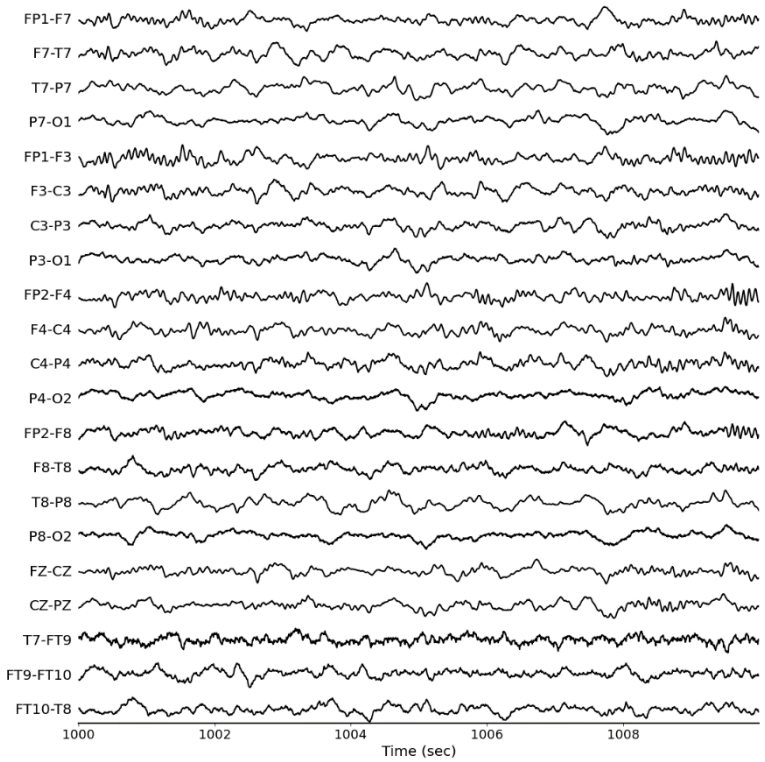


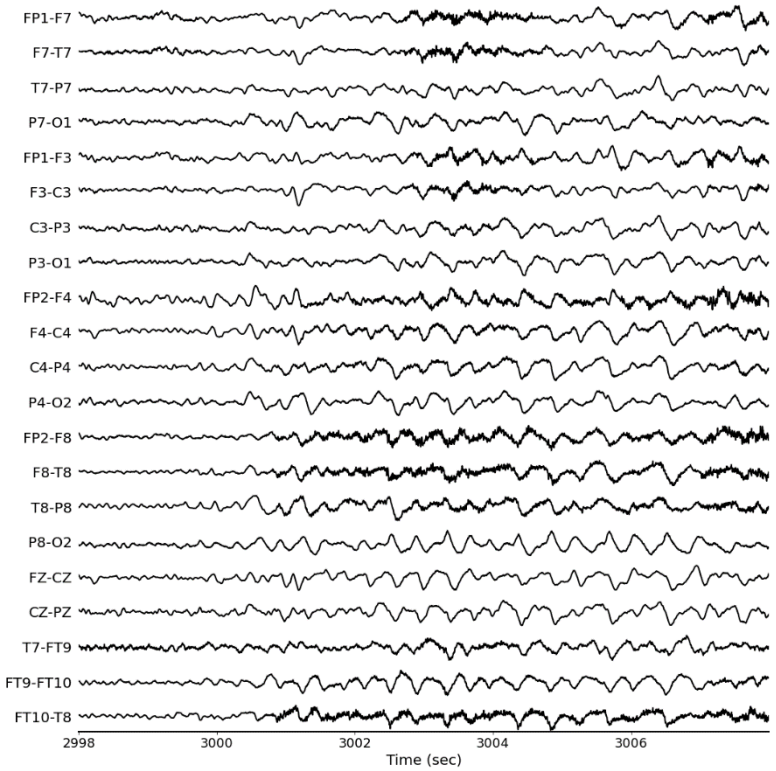


Seizure Patterns in EEG

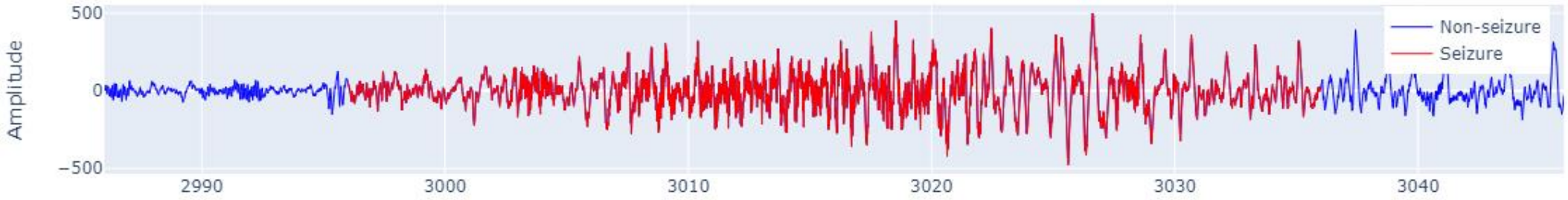


normal

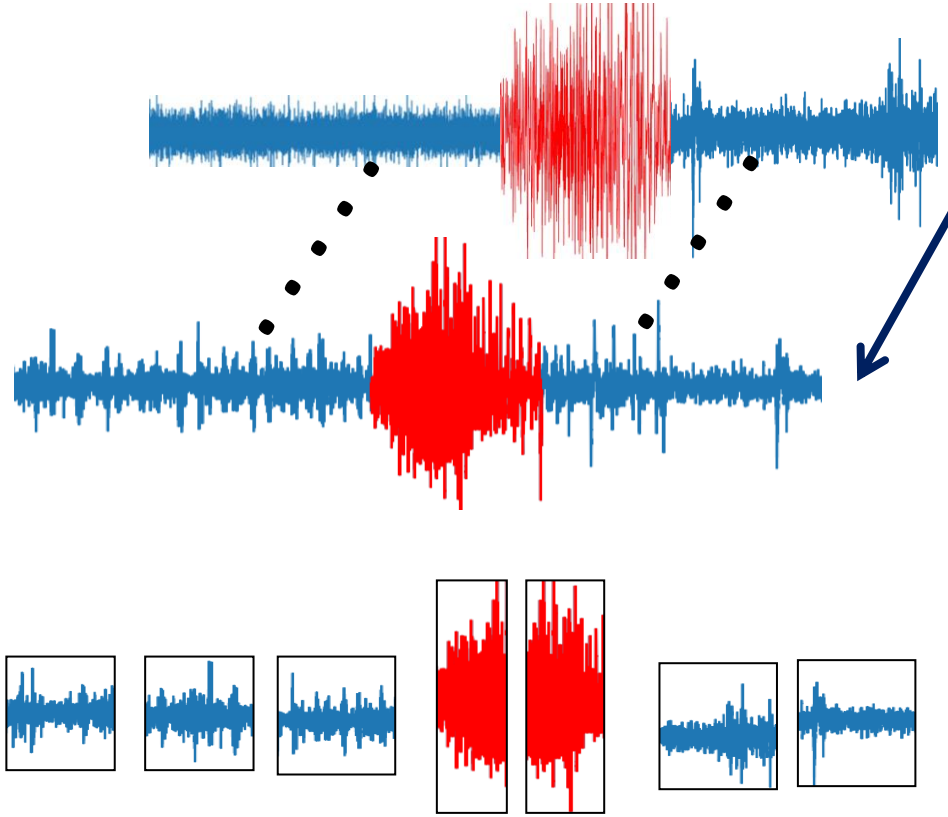




seizure

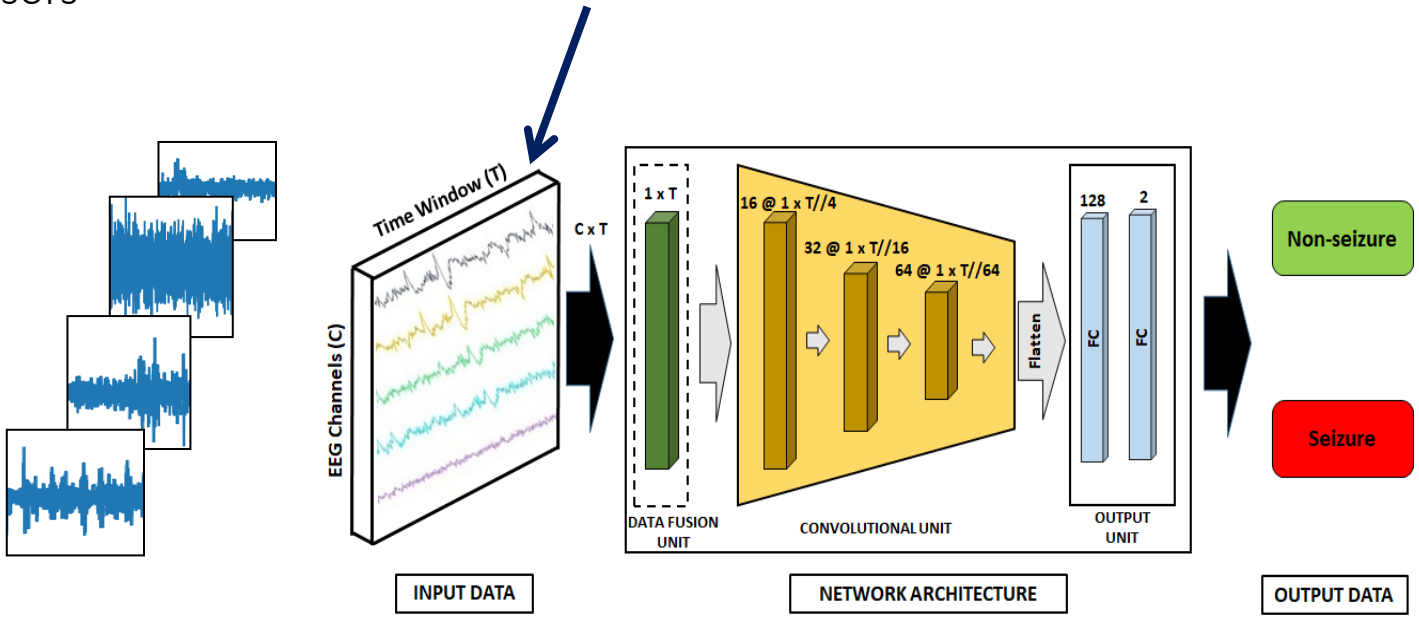


System Pipeline



NCh EEG sensors

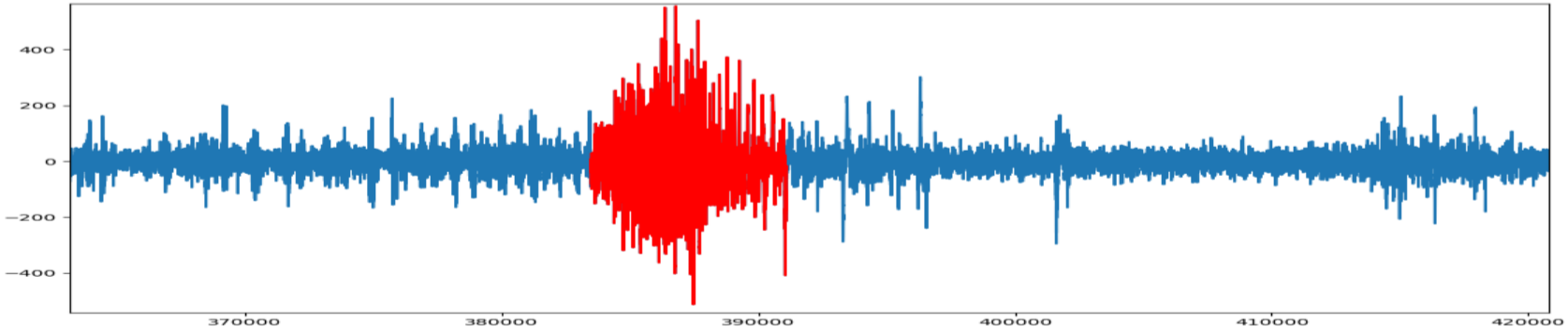
Matrix of dim [NCh, L] for NCh = number of EEG sensors, L =length of a temporal window



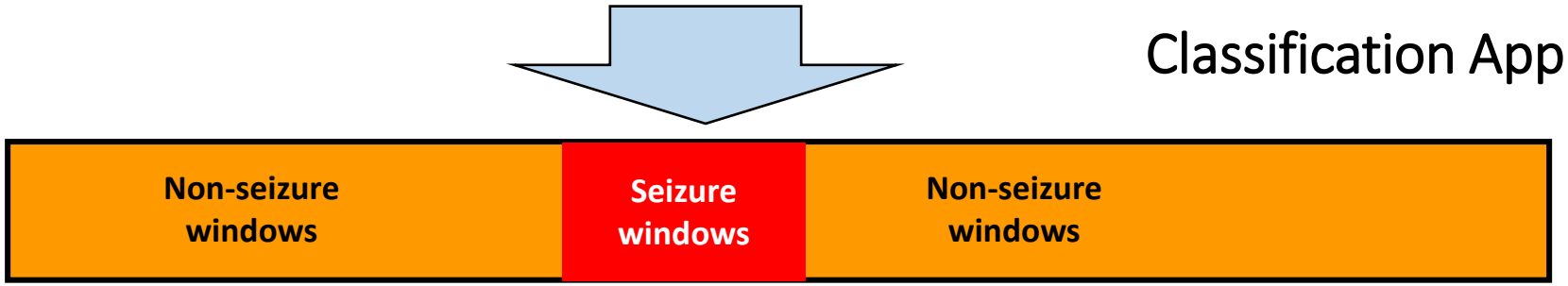
Data Preparation. Extract temporal windows from EEG signals

Window Seizure Detection. Identify seizure presence in Windows from the analysis of NCh EEG signals

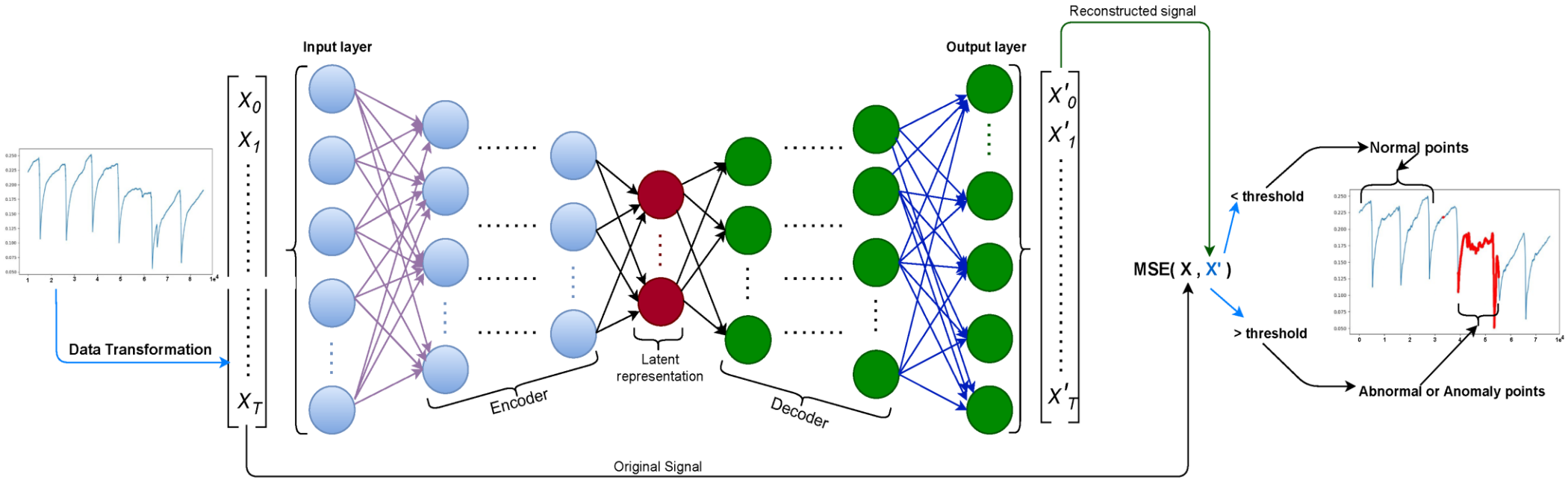
Window Seizure Detection



Classification Approach

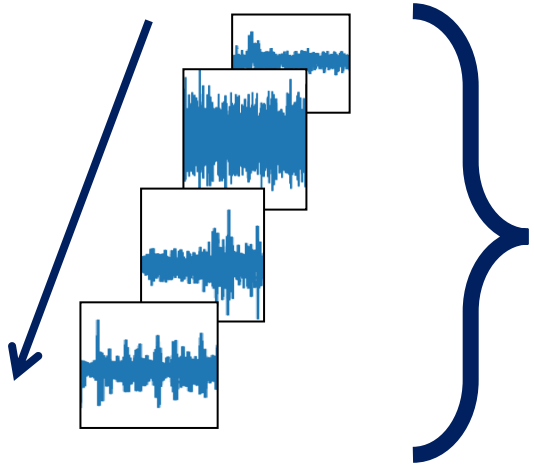


Anomaly Detector



AI Challenges

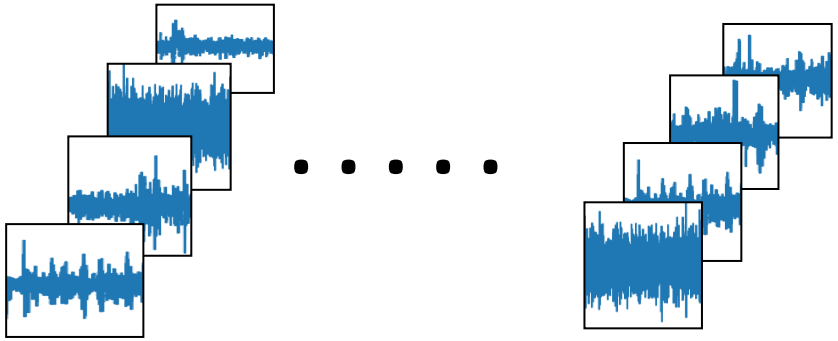
1. How to combine EEG electrode signals (Channel Fusion)



EEG Windows at time T

Representation space of a single window taking into account the information of all NCh EEG channels

2. How to include temporal information of EEG signals



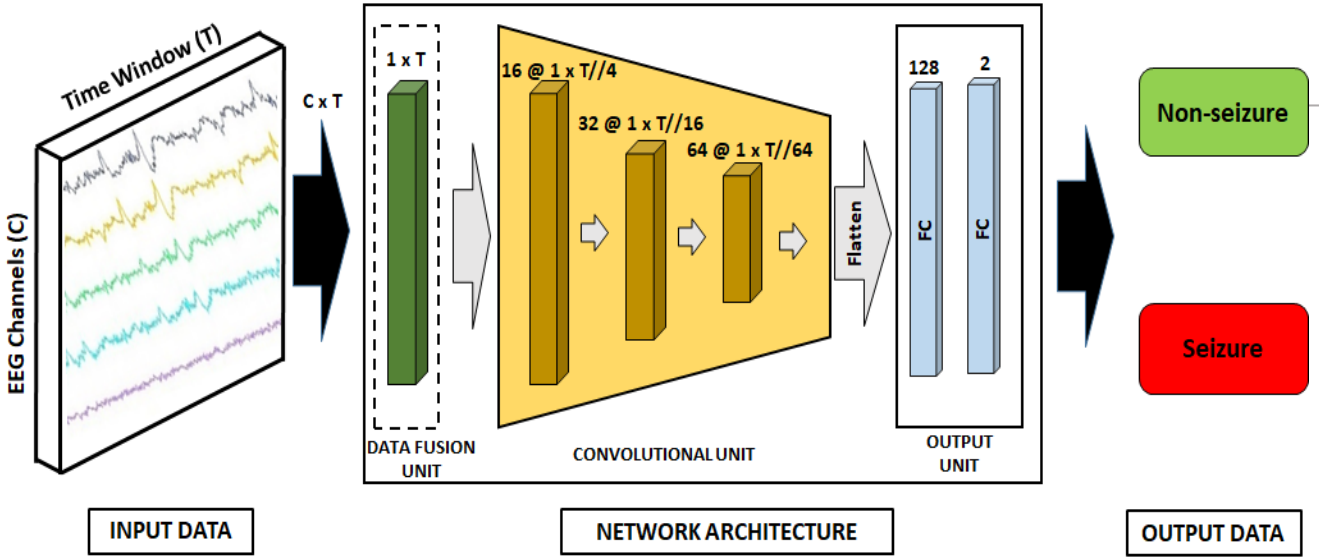
EEG Windows at time T

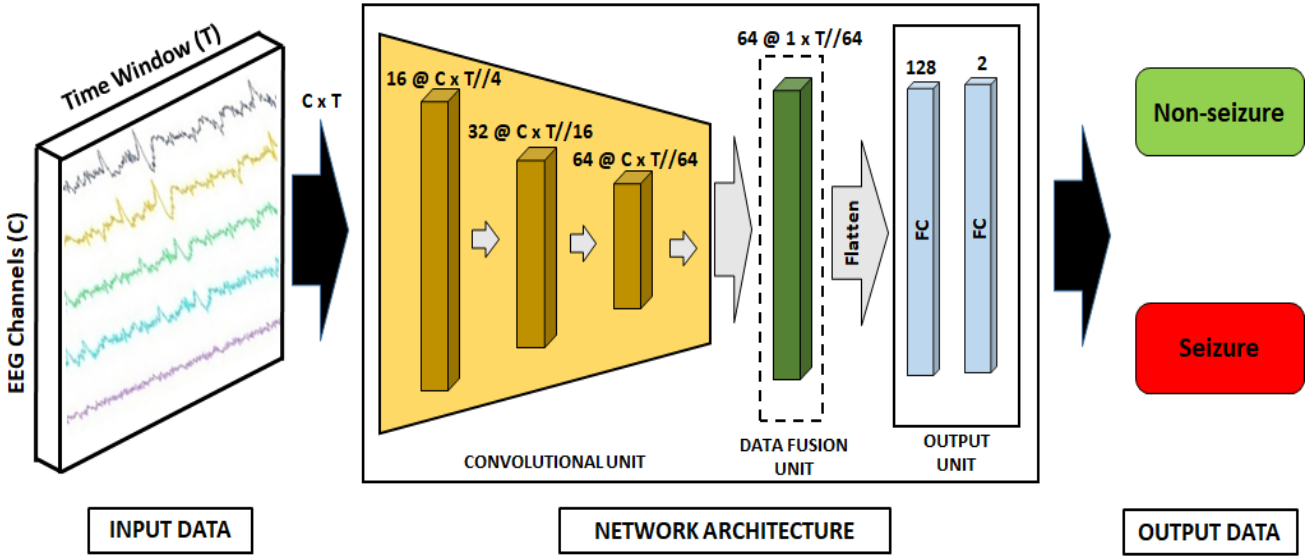
EEG Windows at time T+K

Representation space of a single window taking into account previous Windows in a K-length sequence

Channel Fusion

Fusion of EEG channels at input level (like image CNNs)



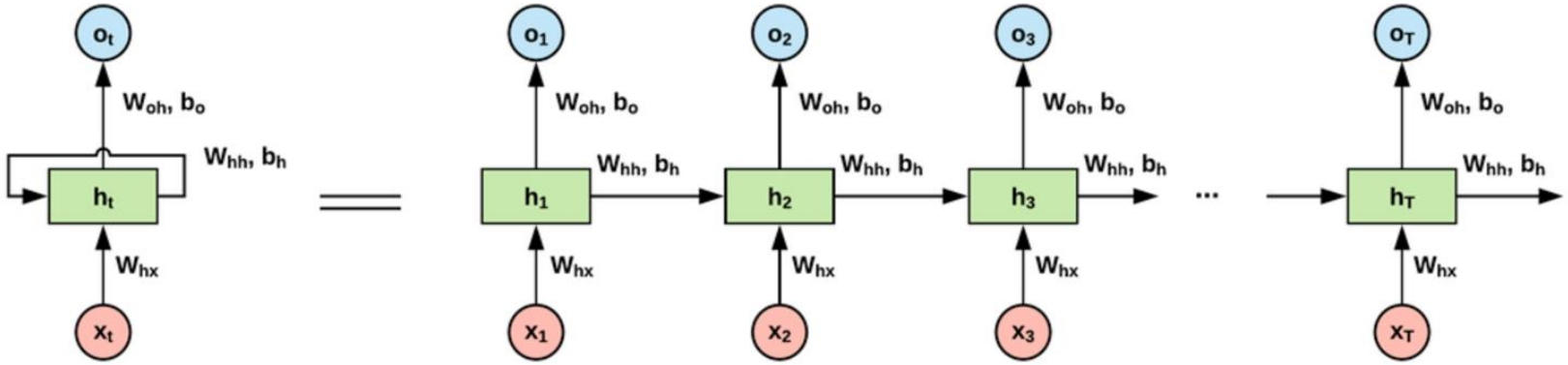


Fusion of EEG channels at feature level (after extracting relevant features for each sensor)

Recognition of the Mental Workloads of Pilots in the Cockpit Using EEG Signals (mdpi.com)

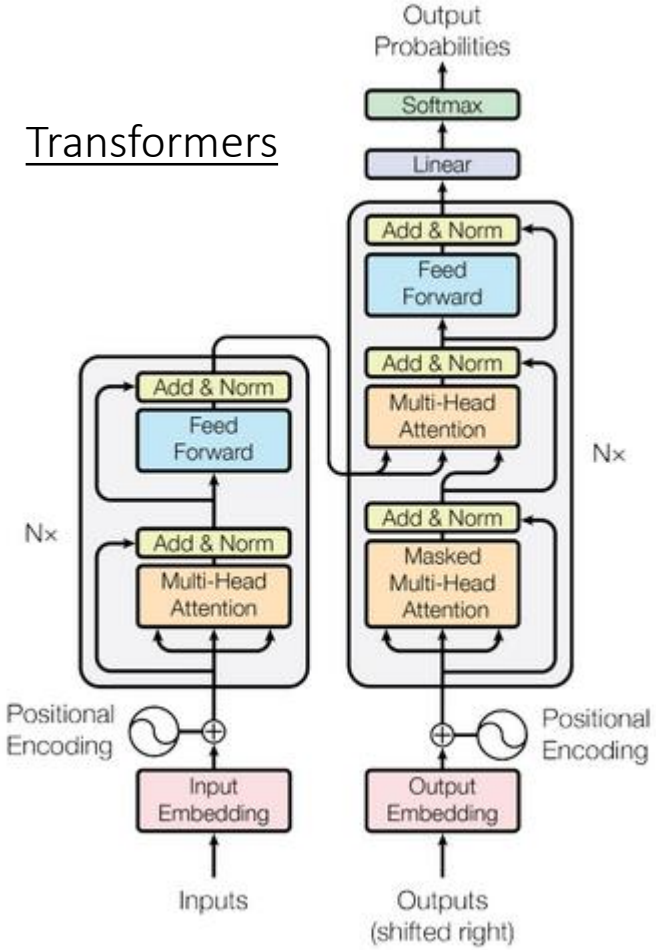
AI Challenges

2. How to include temporal information of EEG signals



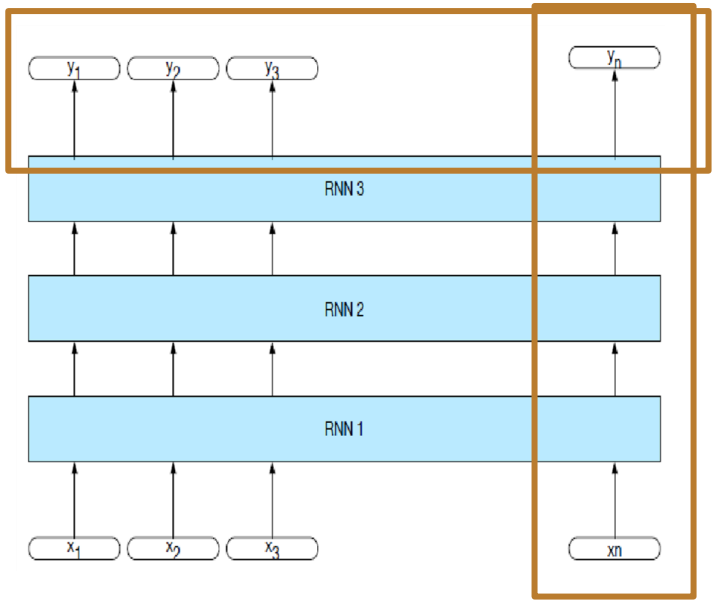
Recurrent Networks (LSTM)

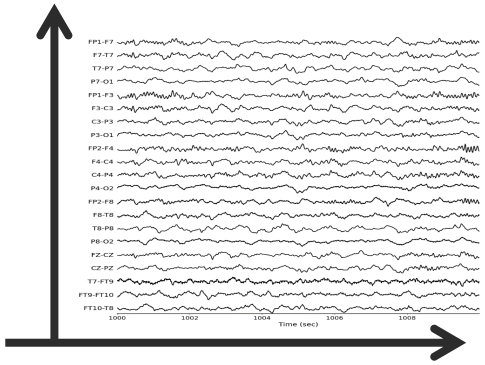
Transformers



Temporal Information

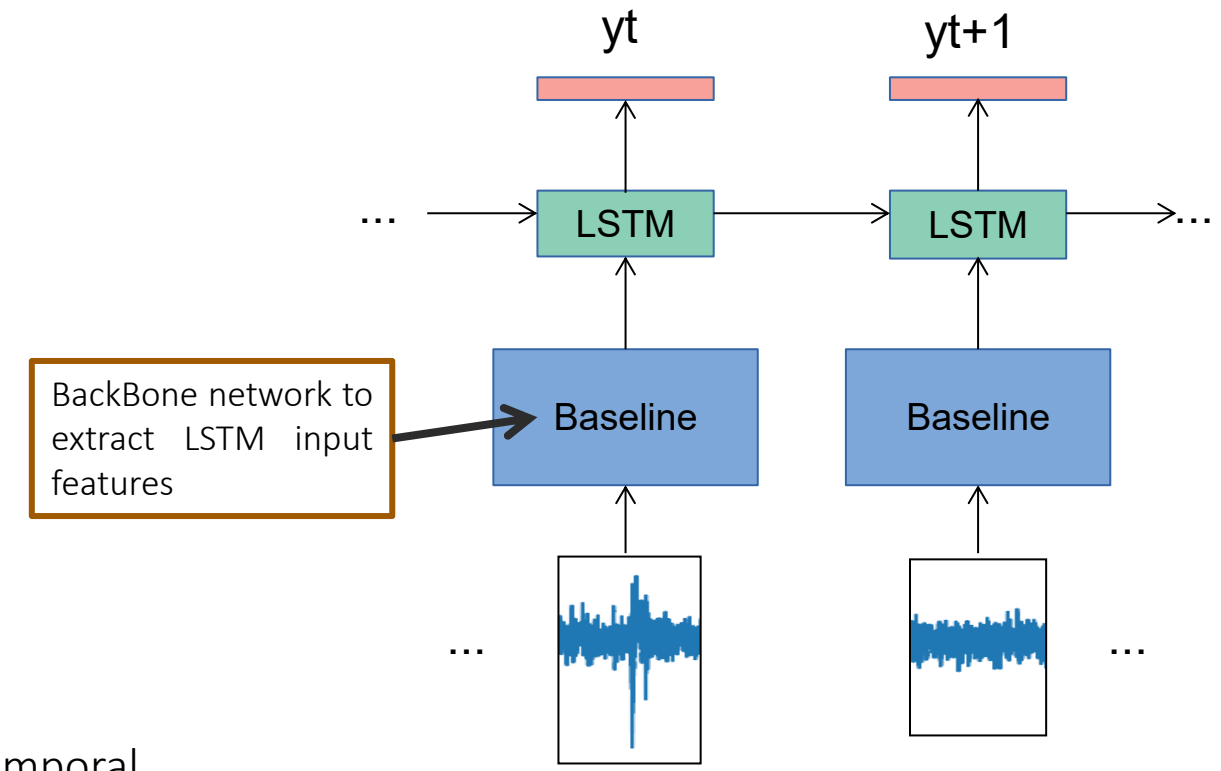
Combine channels using temporal information





X=[21,T] is considered a temporal signal of length T=128 represented by the 21 channels

Process consecutive temporal windows to classify the last one



AI Challenges

3. Personalization level of models

Population Approach

Test: unseen subjects

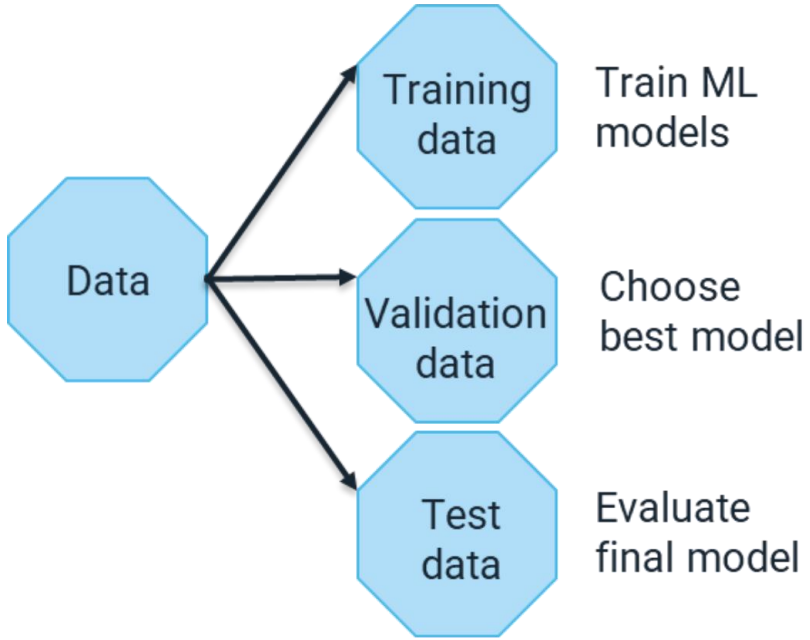
Personalized Approach

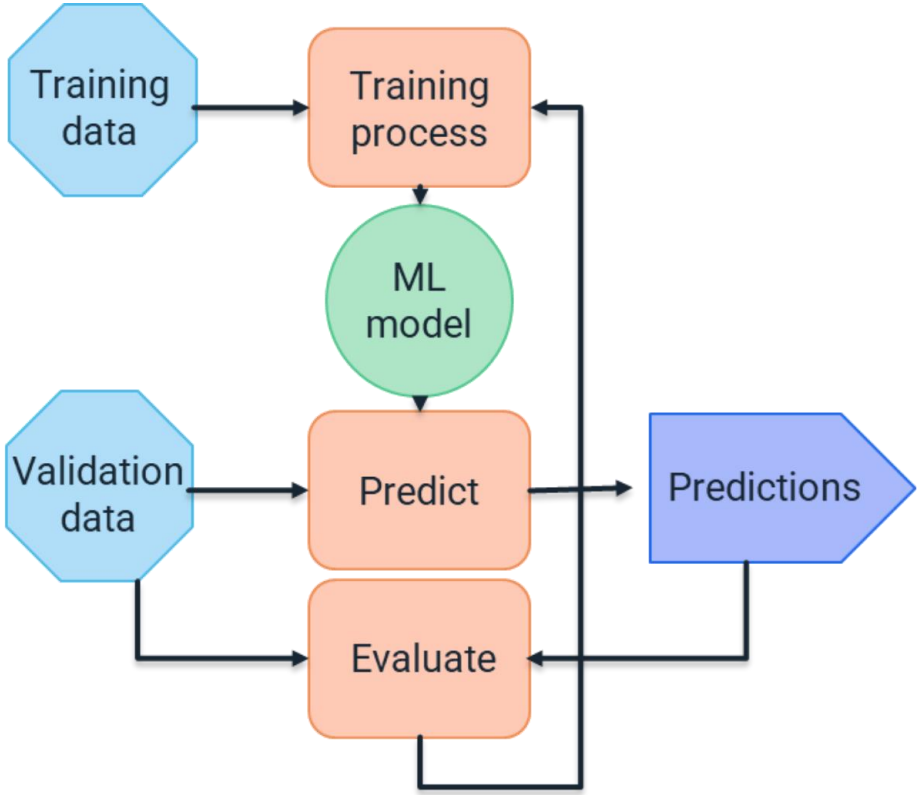
Test: unseen event of same set of subjects

AI Challenges

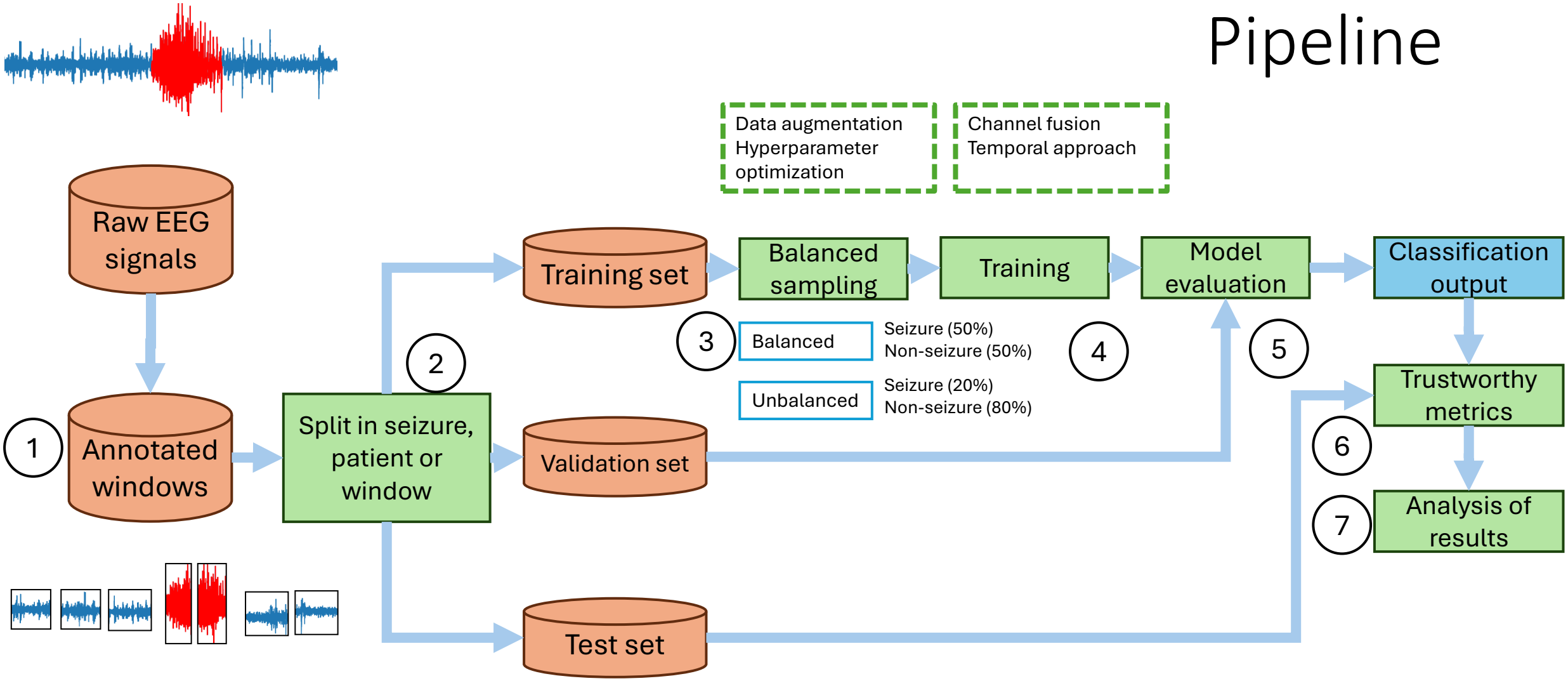
3. Personalization level of models

Dataset split:





Challenge Pipeline



Specific Objectives

1. Implement two systems:

    1.1 Baseline single window classifier with Channel fusion

    1.2 Temporal approach using LSTM

2. Compare performance at different generalization levels:

    2.1 Personalized Model

    2.2 Population Model

Schedule

March

Date	Description	Time
10/2/2026 (Tuesday)	Challenge presentation. Data Base Exploration Materials in virtual campus: Challenge Presentation, CHB-MIT Dataset.	15–17
12/2/2026 (Thursday)	Advanced Preprocessing & Data Cleaning. Handling Missing Data. Channel Standardization.	17–19
17/2/2026 (Tuesday)	Baseline Architecture for Channel Fusion. Materials in virtual campus: Architectures. Channel Fusion.	15–17
19/2/2026 (Thursday)	Baseline Architecture for Channel Fusion. Materials in virtual campus: Architectures. Channel Fusion.	17–19
24/2/2026 (Tuesday)	Train and Test of Baseline Architecture at two different generalization levels. Materials in virtual campus: Validation & Verification.	15-17
26/2/2026 (Thursday)	Project 2: First Defence: Version 1	17-19

Schedule

April

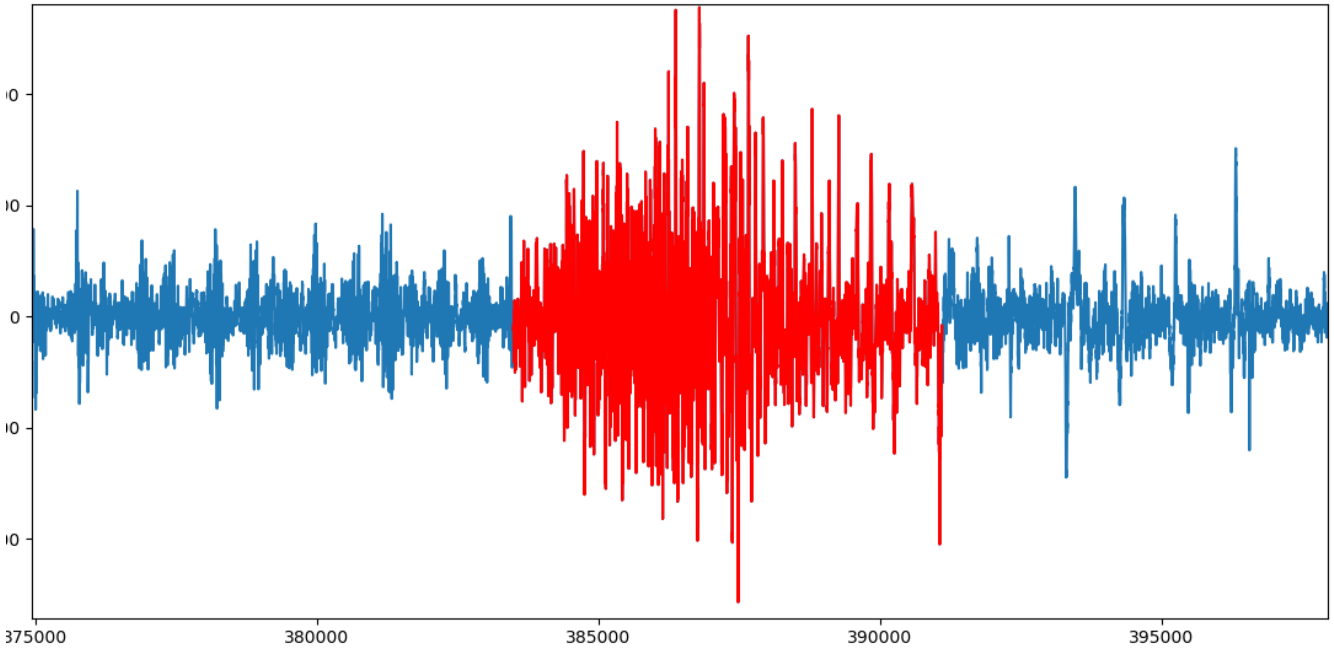
Day	Description	Time
7/4/2026 (Tuesday)	Recurrent Architecture for Processing of Temporal Information. Materials in virtual campus: Architectures. Processing Temporal Data, RNN.	15–17
9/4/2026 (Thursday)	Recurrent Architecture for Channel Fusion. Materials in virtual campus: LSTM, Example Code (ChakrabartiChannelFusion).zip in Processing Temporal Data.	17–19
14/4/2026 (Tuesday)	Recurrent Architecture for Channel Fusion. Materials in virtual campus: LSTM ExampleCode (ChakrabartiChannelFusion).zip in Processing Temporal Data.	15–17
16/4/2026 (Thursday)	Recurrent Architecture for Channel Fusion. Materials in virtual campus: LSTM, Example Code (ChakrabartiChannelFusion).zip in Processing Temporal Data.	17–19
21/4/2026 (Tuesday)	Project 2: Final defense	15-17
28/4/2026 (Thursday)	Project 1: Recovery of Challenge / Project 2: Recovery of Challenge	15-19

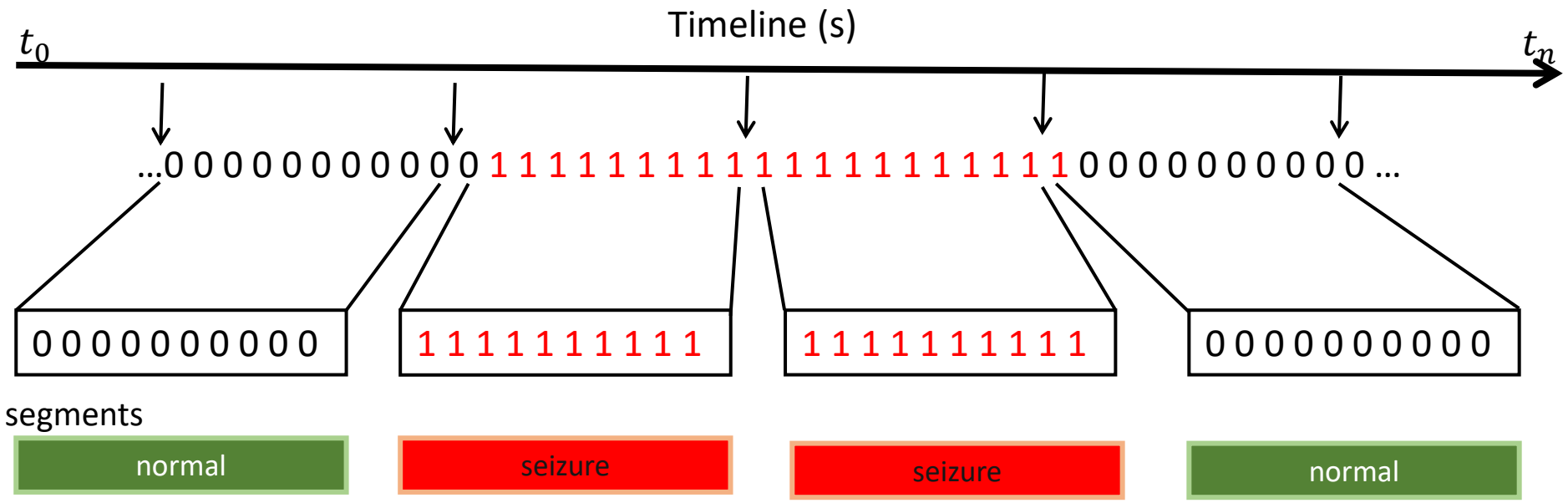
Dataset

CHB-MIT Dataset

Recordings at 256 Hz from 24 subjects with several seizures. Seizure start-end time are annotated.

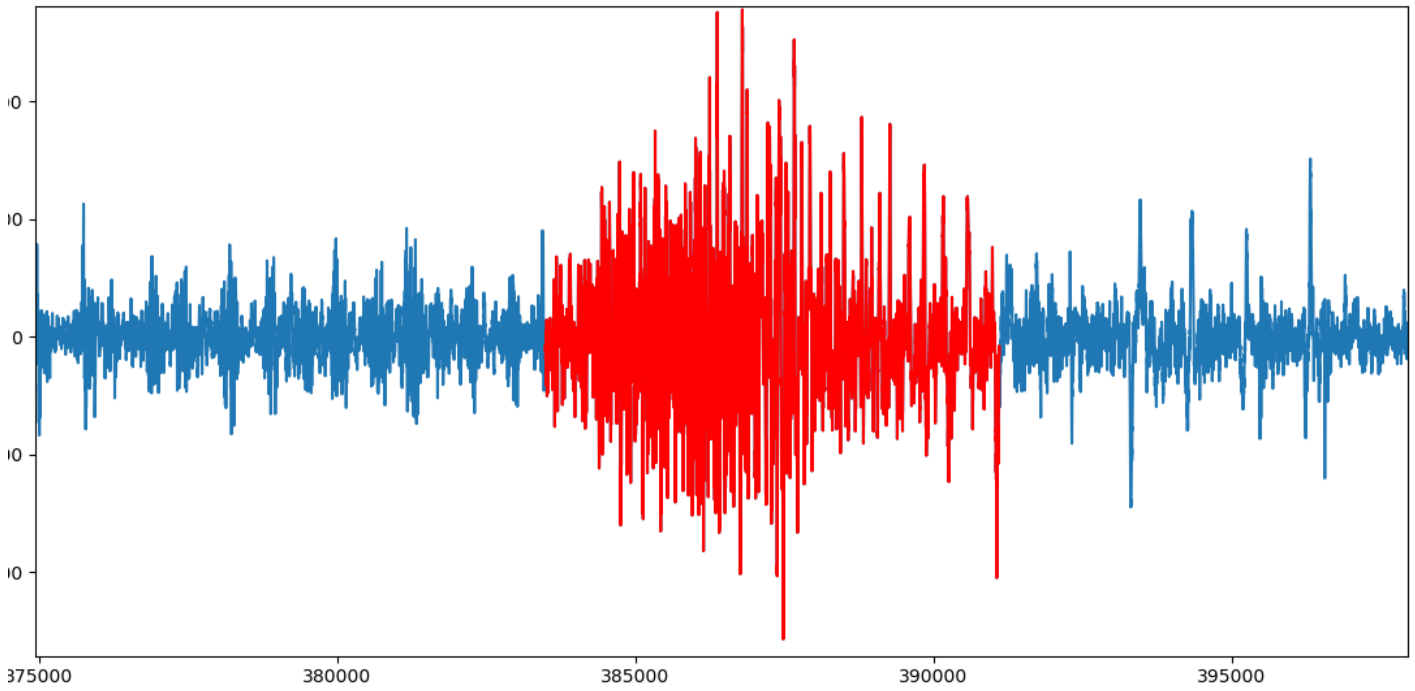
EEG cap with 21 electrodes

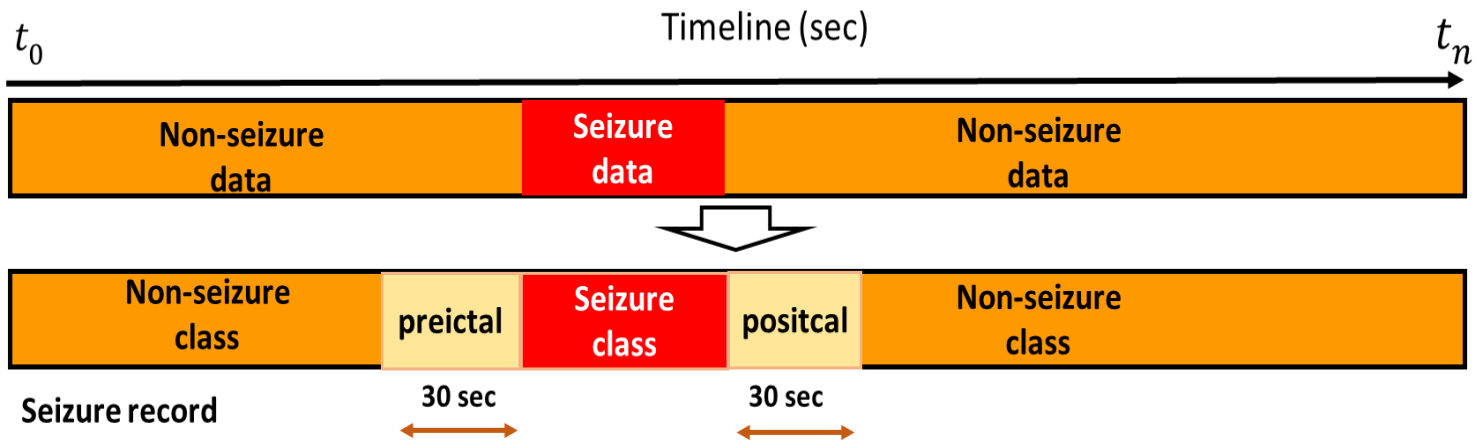




Annotated windows

Signals down-sampled at 128 Hz and cropped in 1 second temporal windows



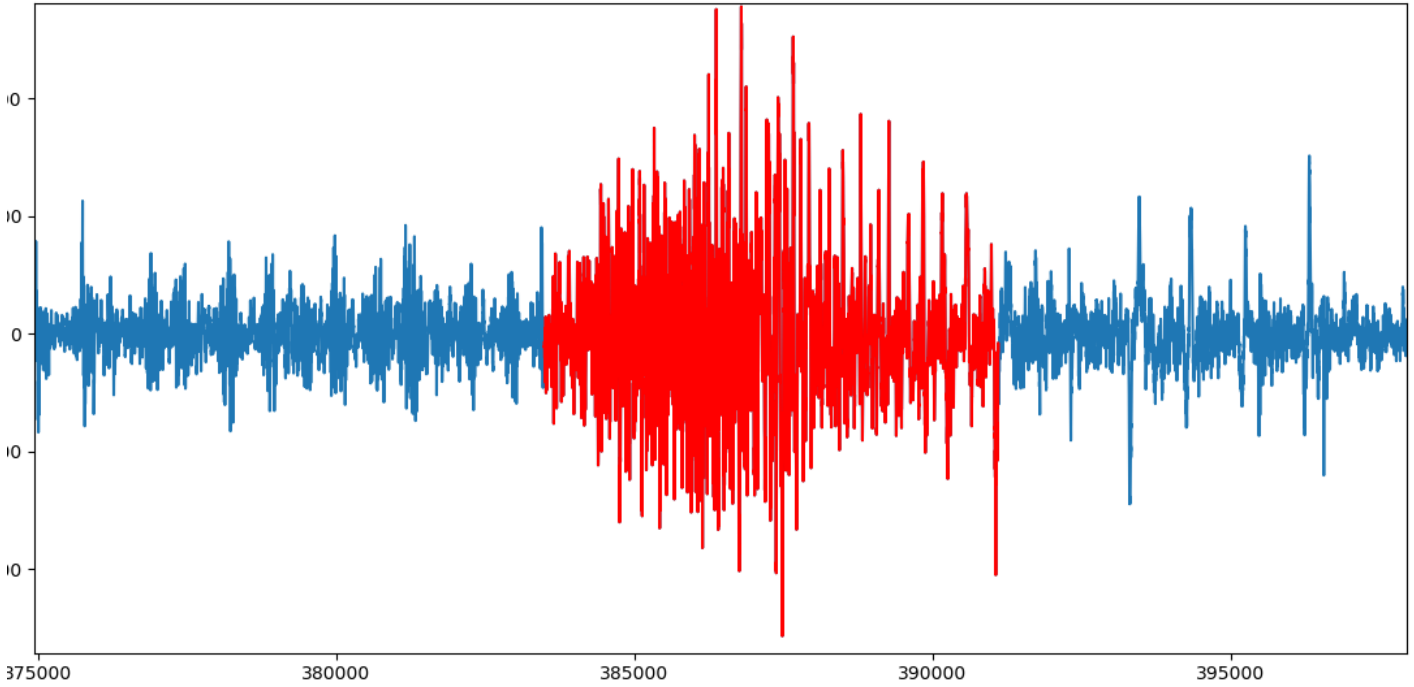


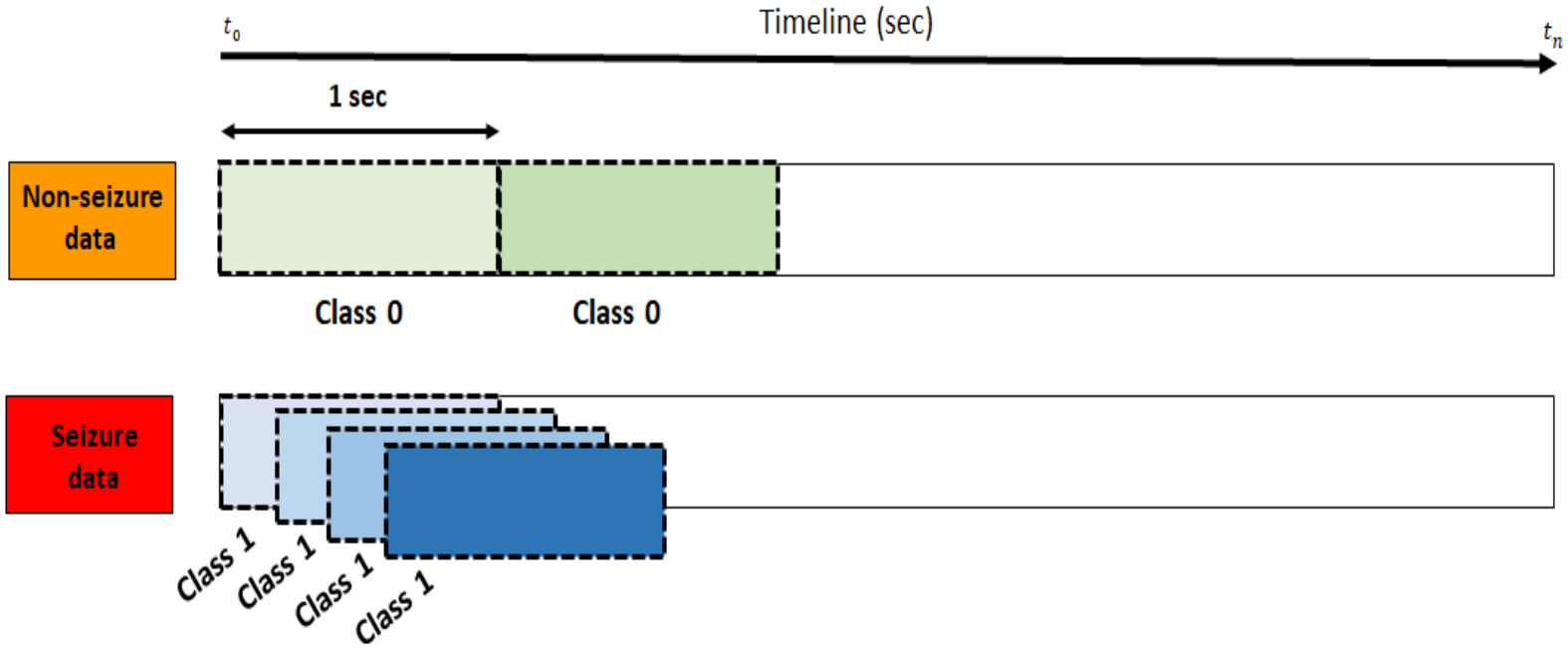
Windows annotations:
0-normal
1-ictal

pre and post-ictal intervals should be discarded

Annotated windows

Ictal windows with 80% overlap for ictal data augmentation





Annotated windows files

1. npz files with EEG Windows signals:

PatID_seizure_EEGwindow_1.npz
PatID ={chb_01, …., chb24}
EEG_win is ndarray of size [NWinPatID, 21, 128]

Number of annotated windows

Number of EEG Channels

Length of the 1 sec. window

Annotated windows files

2. parquet with EEG Windows Metadata: PatID_seizure_metadata_1.parquet

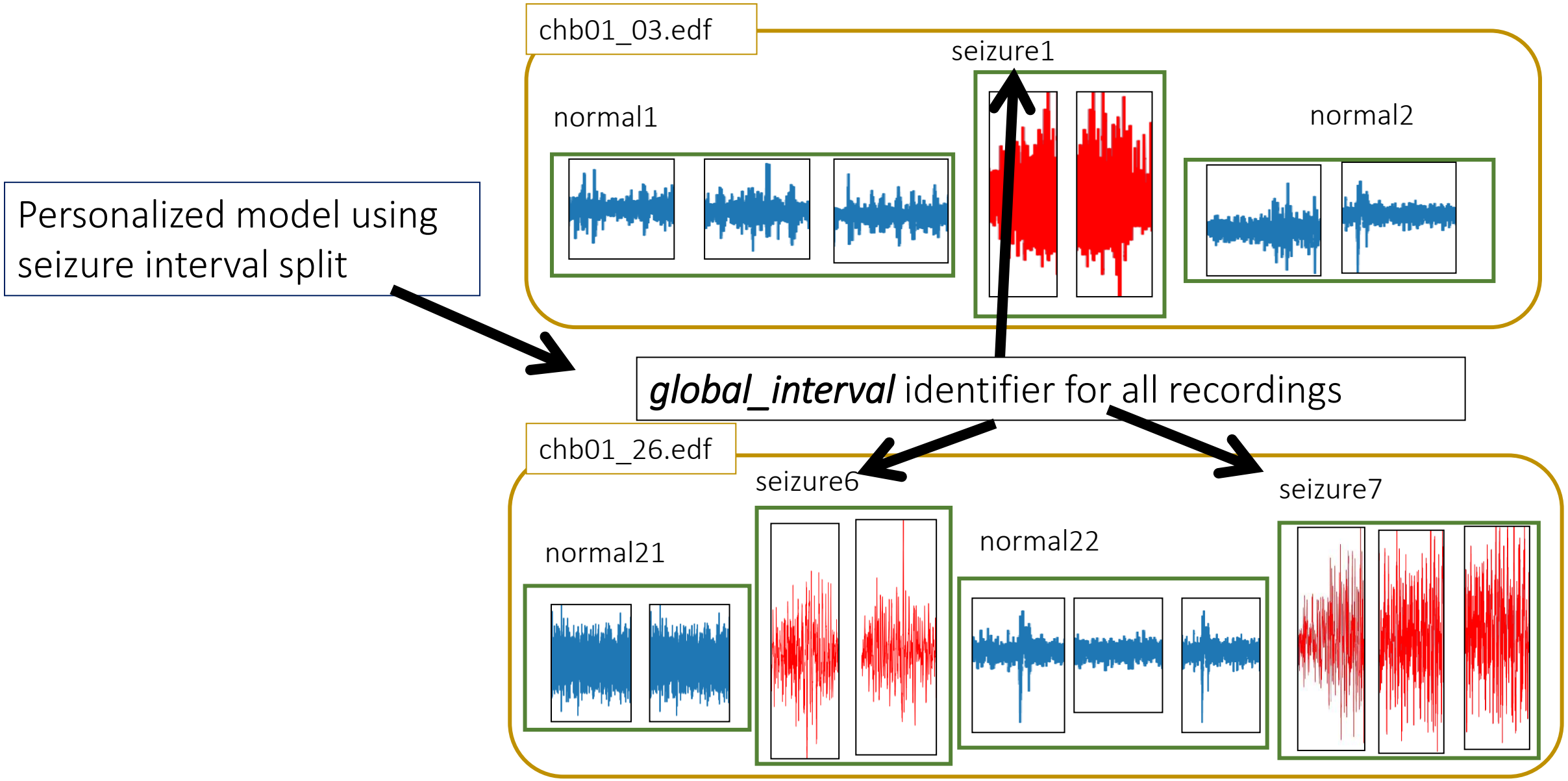
Dataframe with class, recording identifier (.edf), normal/ictal Interval identifiers of each window

	class	filename_interval	global_interval	filename
0	0	1	1	chb01_03.edf
1	0	1	1	chb01_03.edf
2	0	1	1	chb01_03.edf
…	…	…	…	
26525	1	2	7	chb01_26.edf
26526	1	2	7	chb01_26.edf
26527	1	2	7	chb01_26.edf
26528	1	2	7	chb01_26.edf

Usefull for data split at both levels of personalization

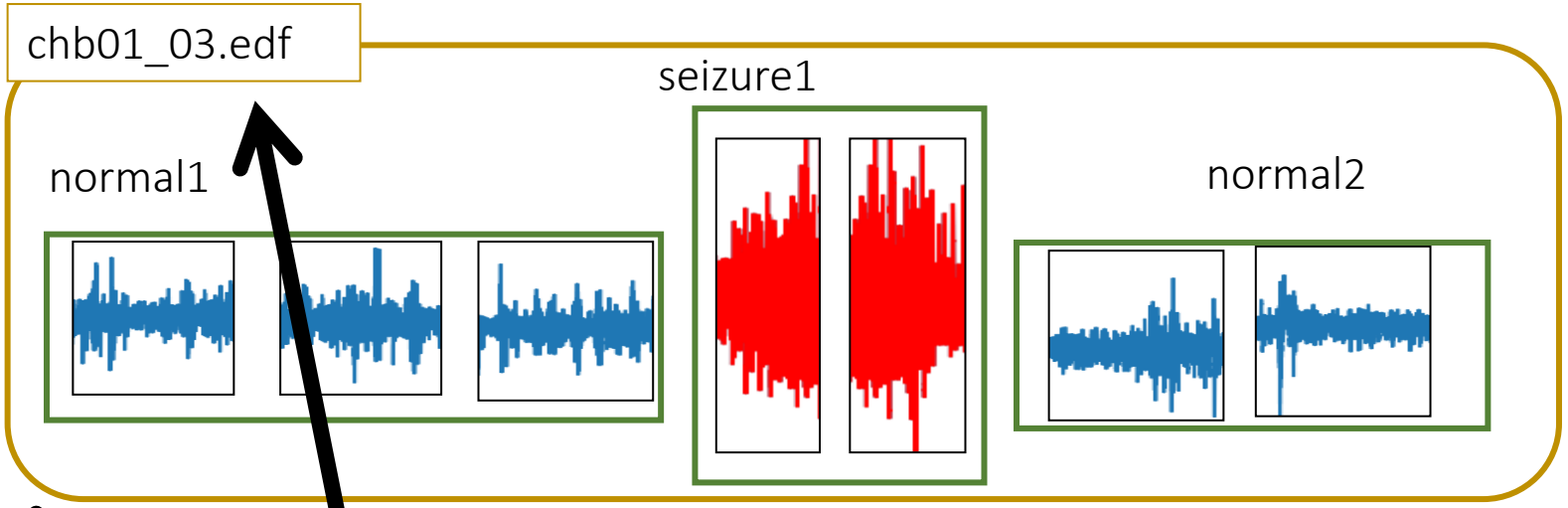
```
metadata=pd.read_parquet
(file,engine='fastparquet')
```

Annotated windows metadata

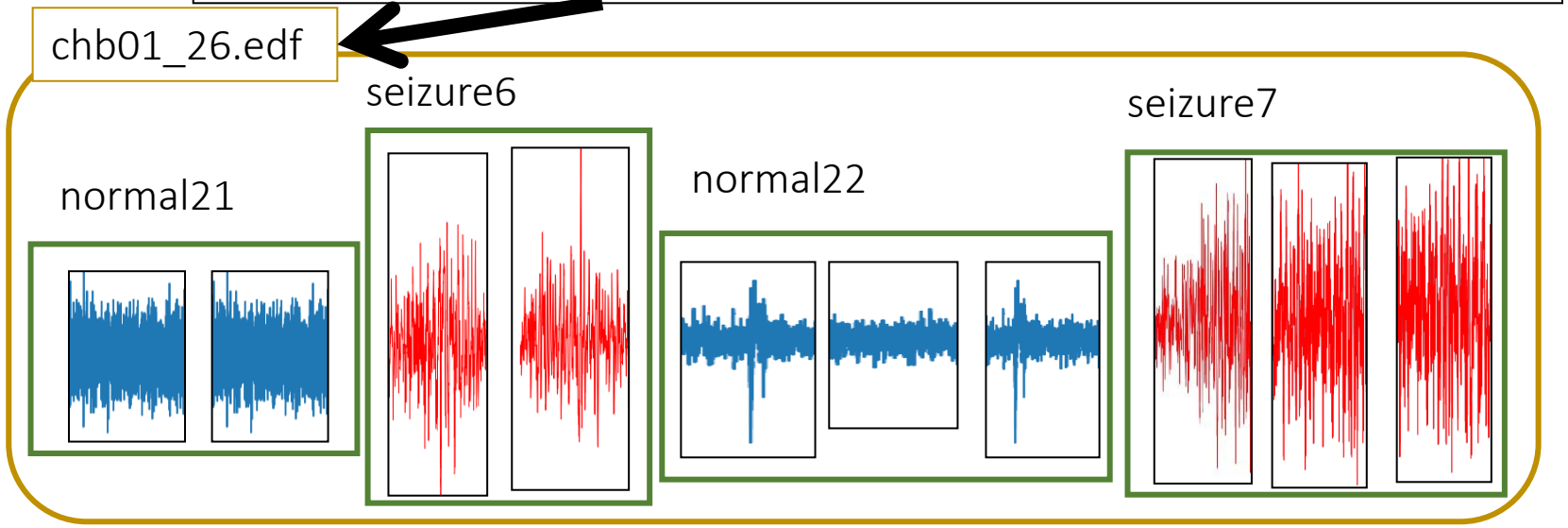


Annotated windows metadata

Personalized model using patient recording split



filename identifies different recordings of the patient



Annotated windows metadata

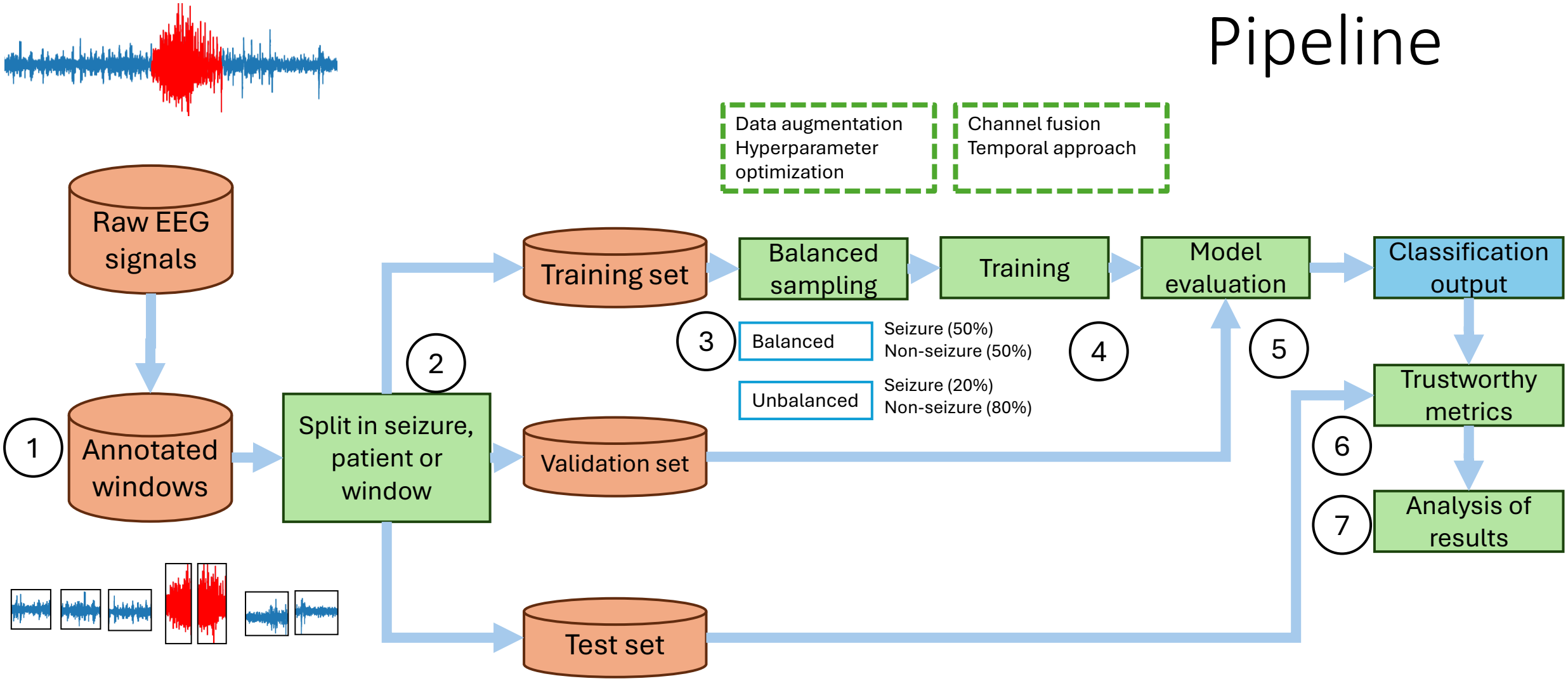
Population model using patient split (leave-one-out)

filename.Split('_')[0] identifies the patient

	class	filename_interval	global_interval	filename
0	0	1	1	chb01_03.edf
1	0	1	1	chb01_03.edf
2	0	1	1	chb01_03.edf
…	…	…	…	
26525	1	2	7	chb01_26.edf
26526	1	2	7	chb01_26.edf
26527	1	2	7	chb01_26.edf
26528	1	2	7	chb01_26.edf

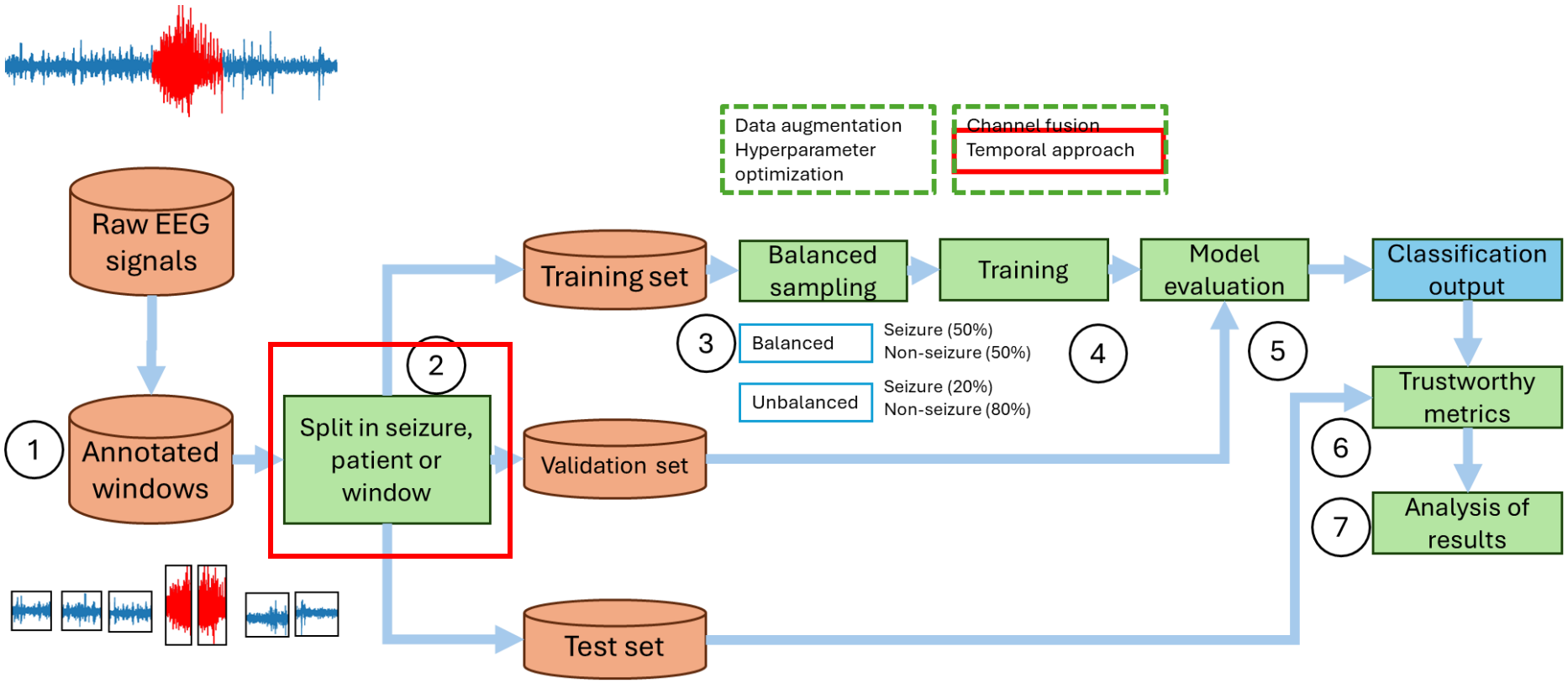
Objectives

Challenge Pipeline



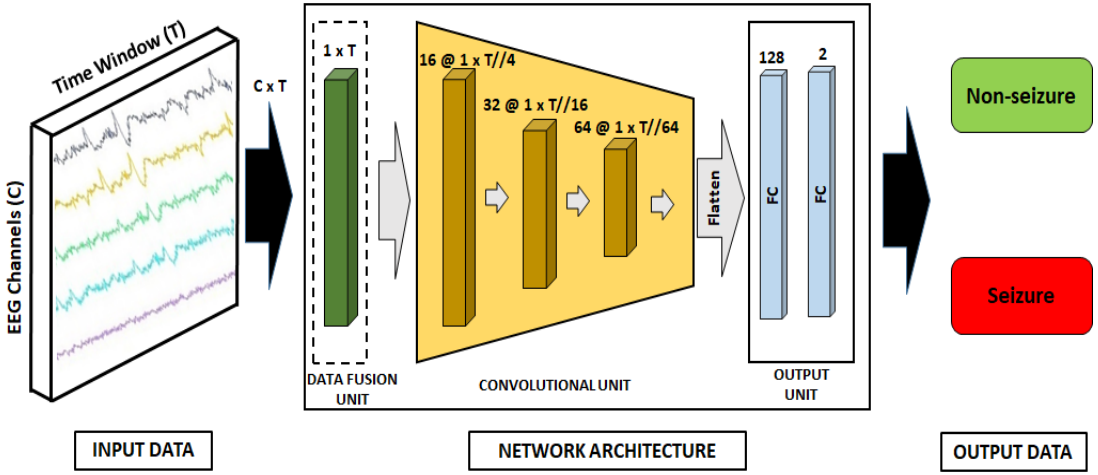
Challenge objectives

1. Baseline static (classification) approach (channel fusion).
2. Approach including temporal information
3. Evaluate methods' generalization

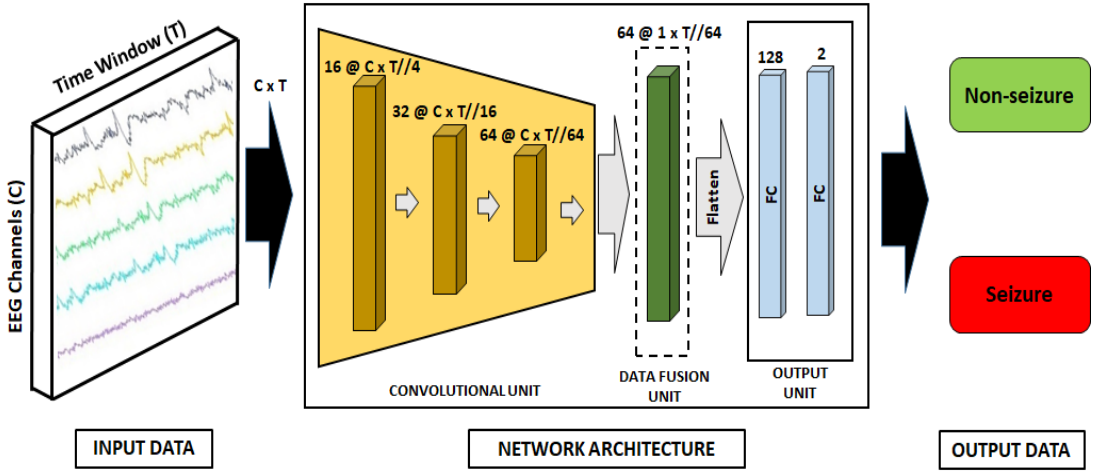


Baseline architecture

Binary Classifier combining channels at either input or feature levels

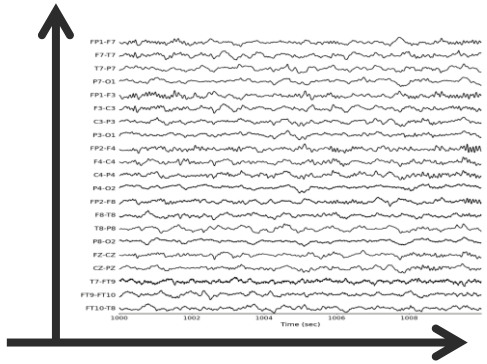


Fusion of EEG channels at input level

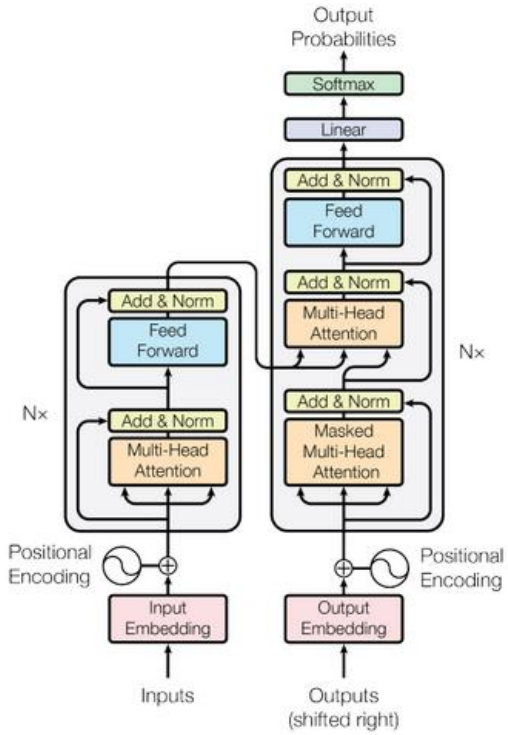


Fusion of EEG channels at output level

Temporal Information

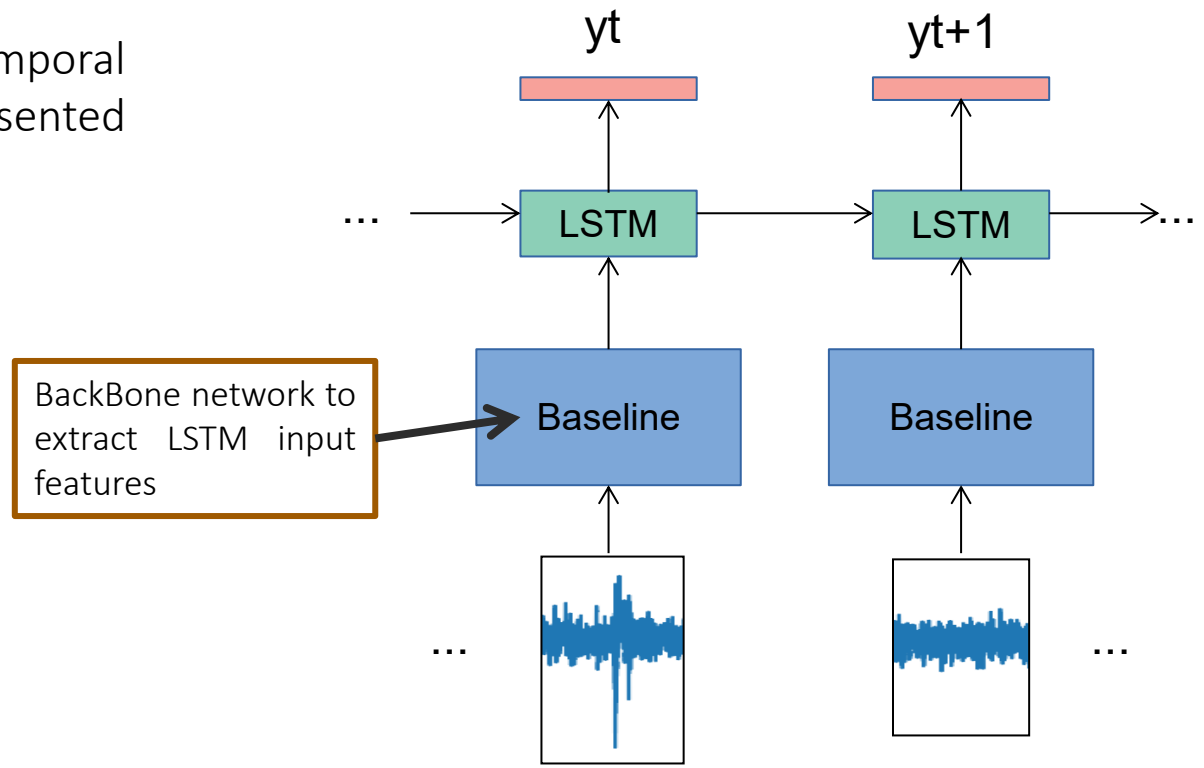


X=[21,T] is considered a temporal signal of length T=128 represented by the 21 channels

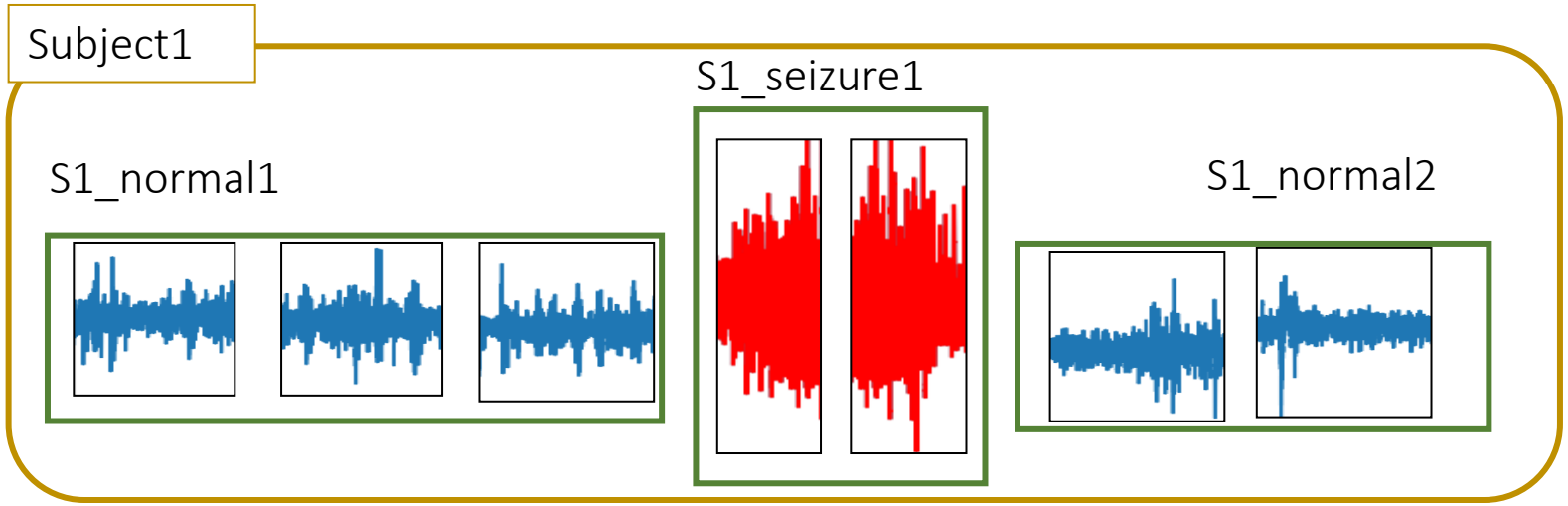


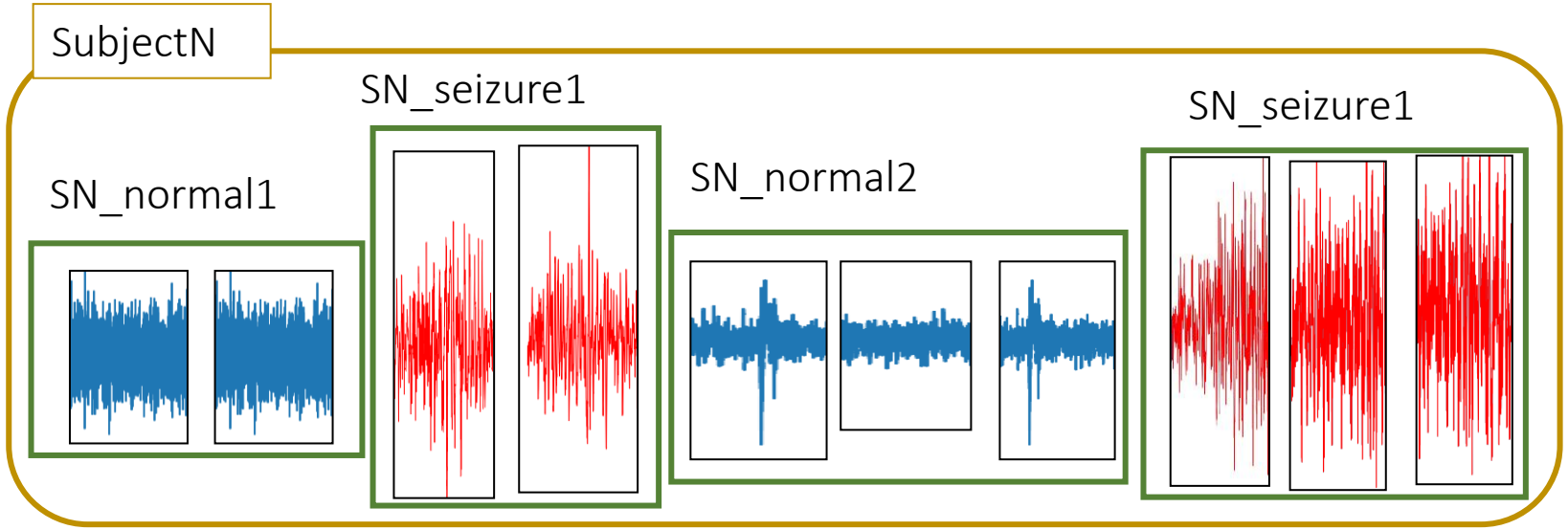
Transformers

Process consecutive temporal windows to classify the last one



Evaluation



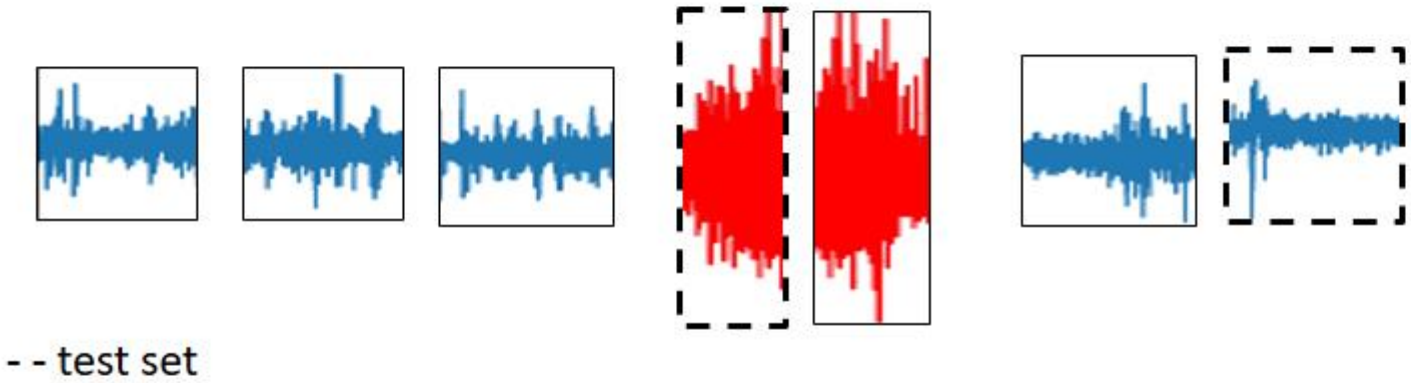


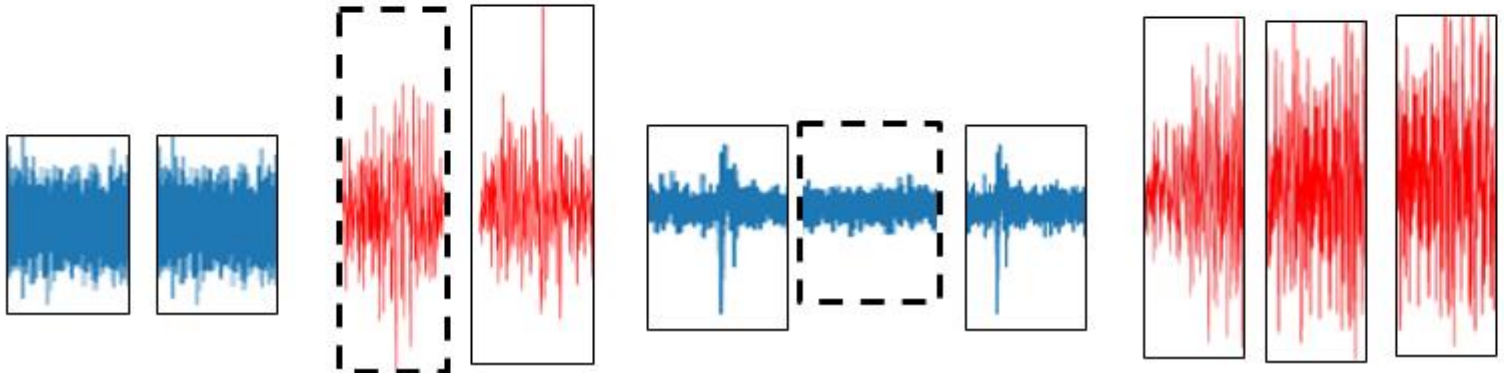
Sampling Levels:

1. Window
2. Seizure
3. Subject

Window Level

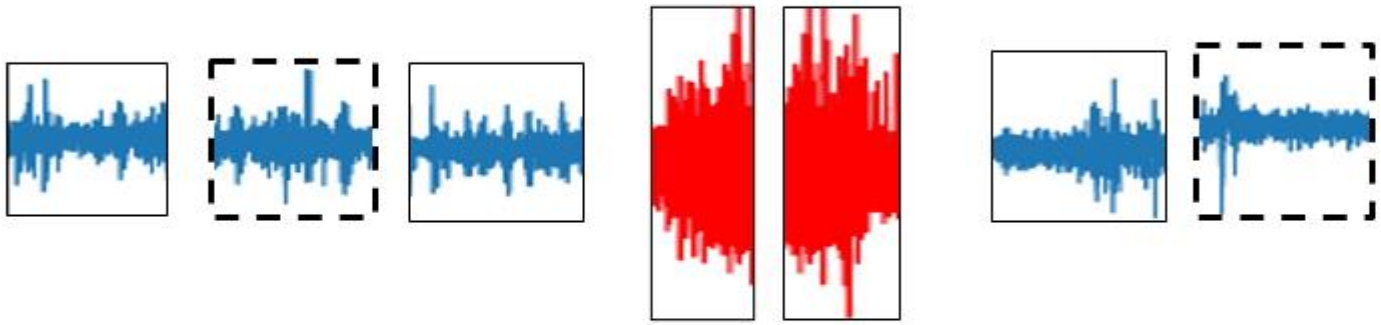
1. **Data Split:** K-fold on windows (experimental sampling unit)



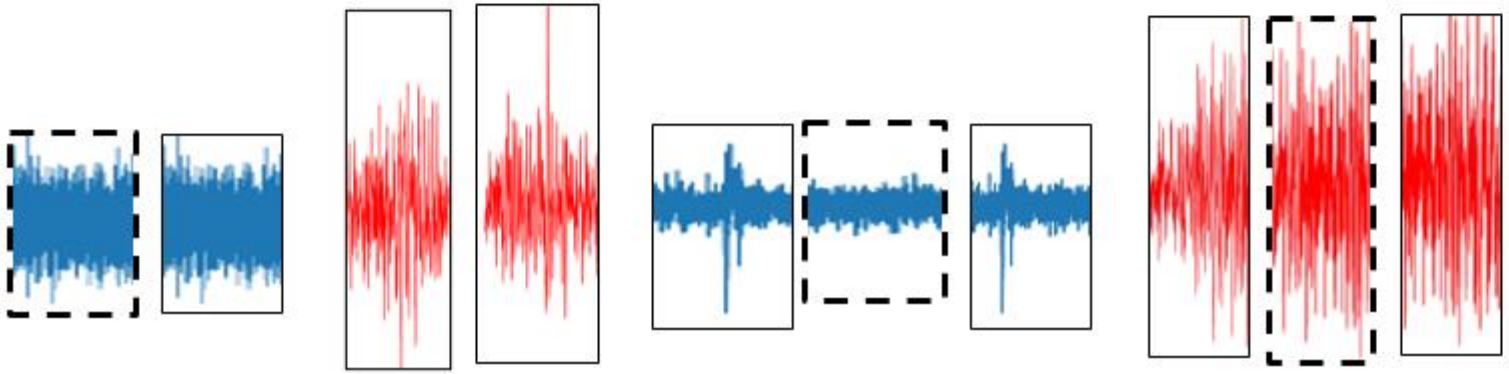


Window Level

1. **Data Split:** K-fold on windows (experimental sampling unit)

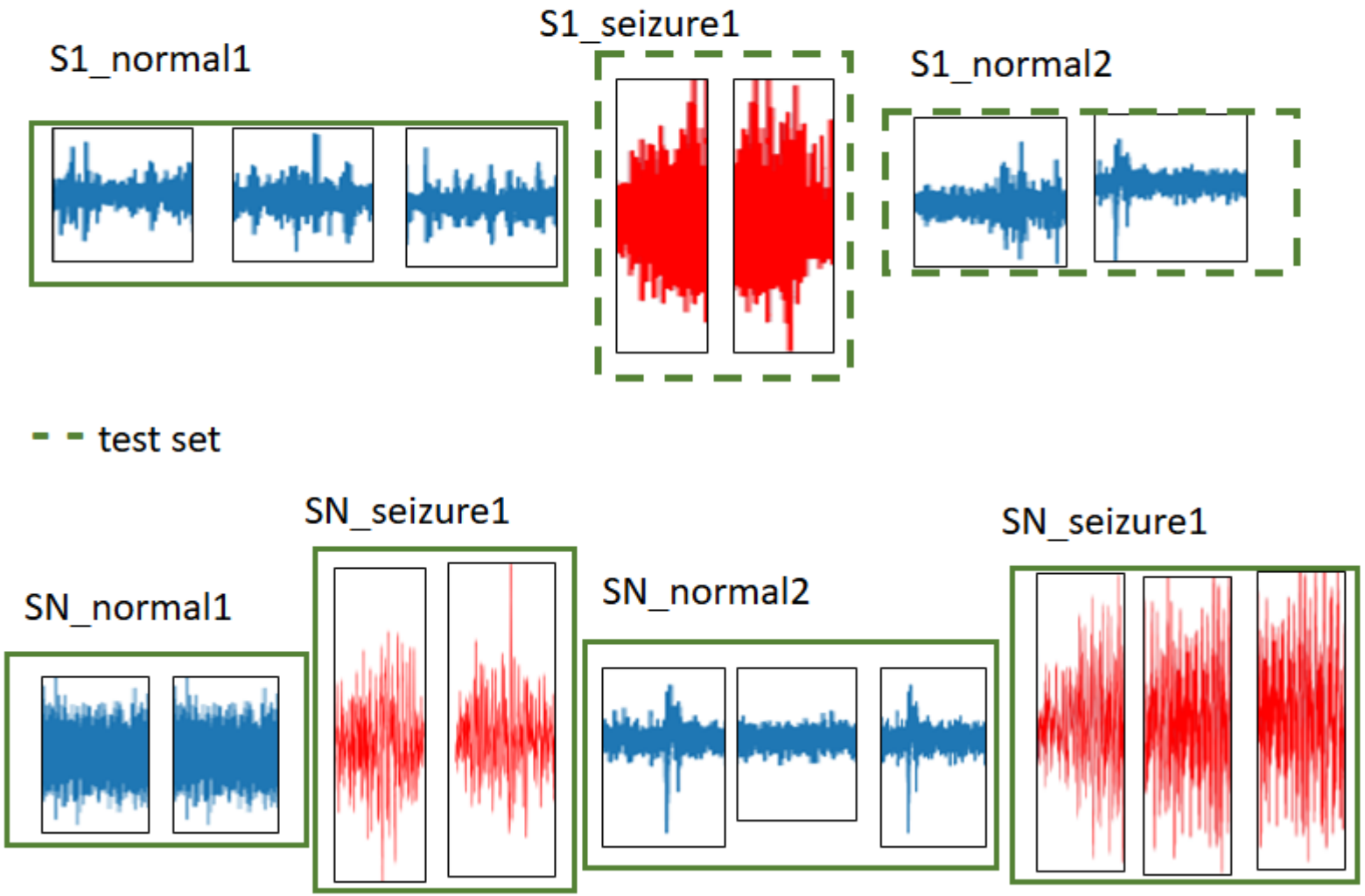


- - test set



Seizure Level

2. Data Split: K-fold on seizure/normal episodes (leave-one-seizure-out)



Seizure Level

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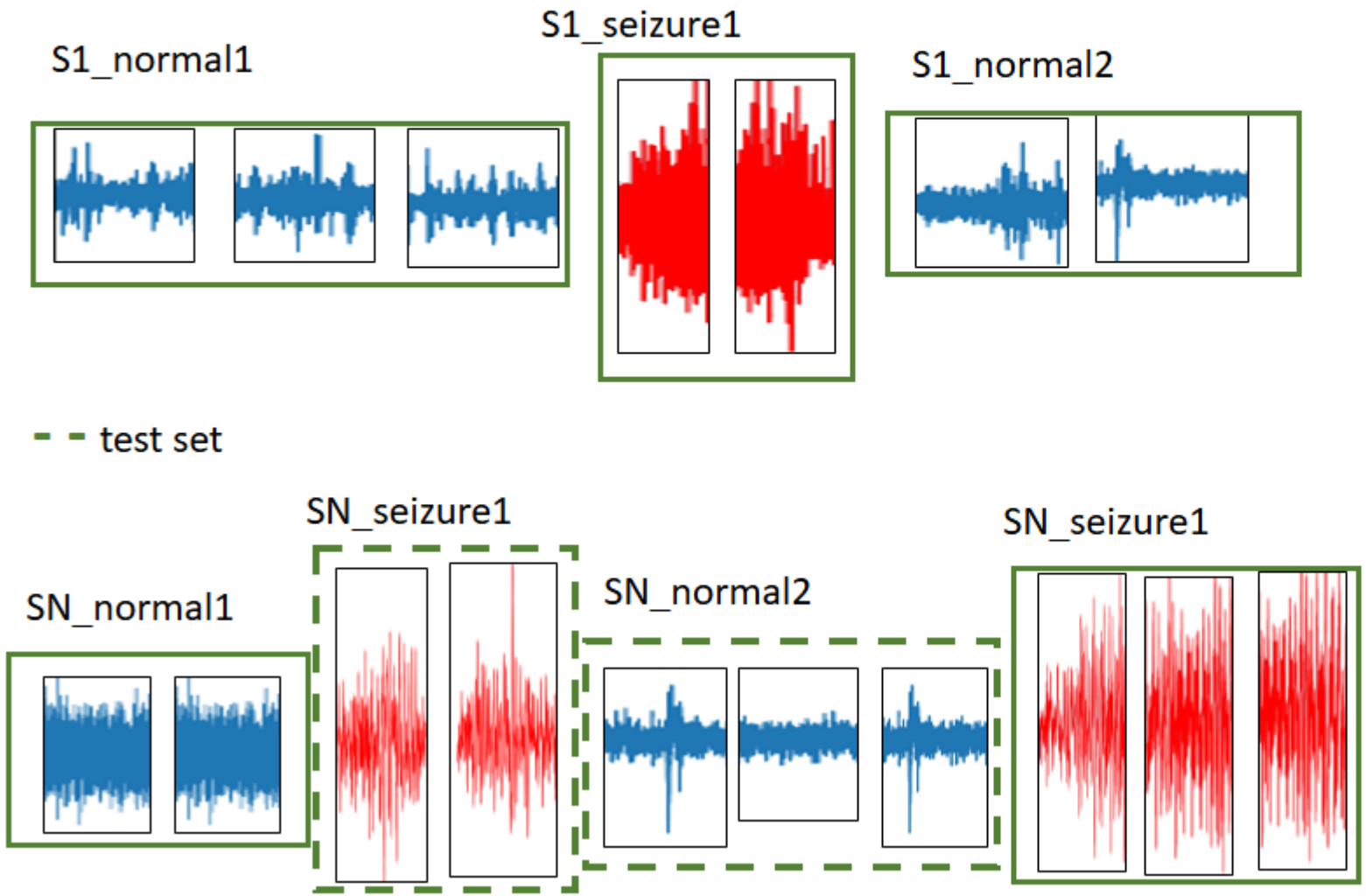

Seizure Level

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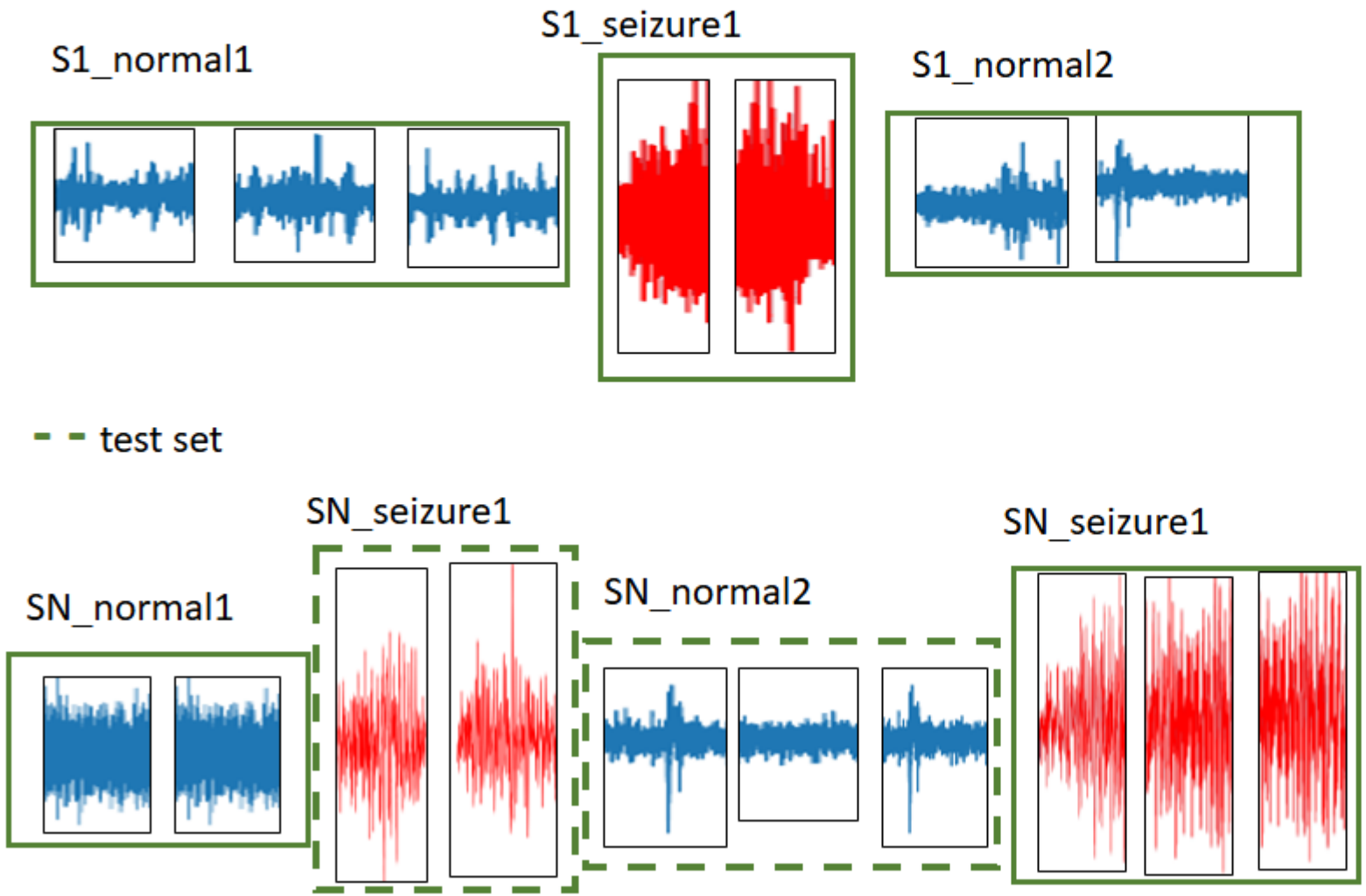
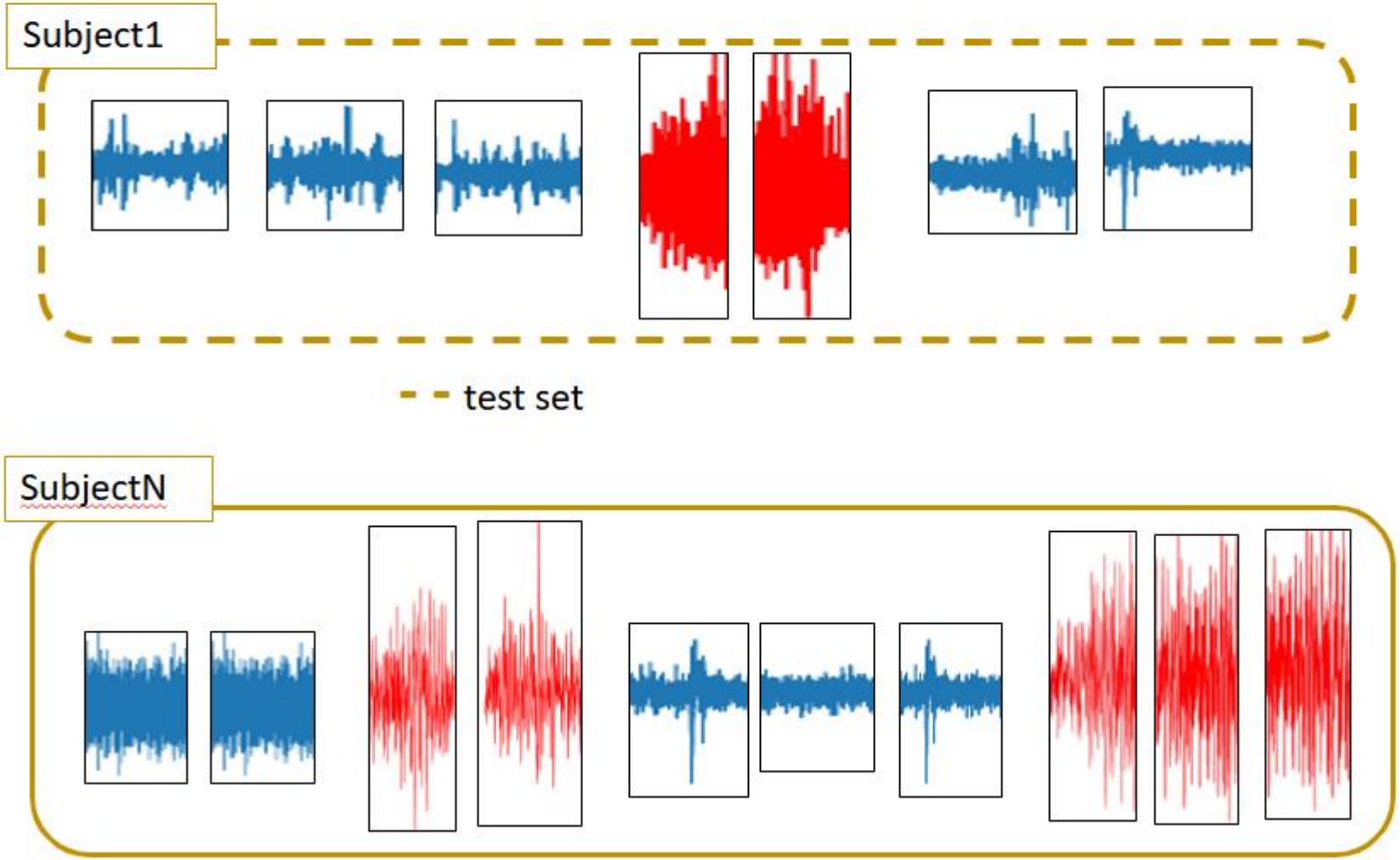


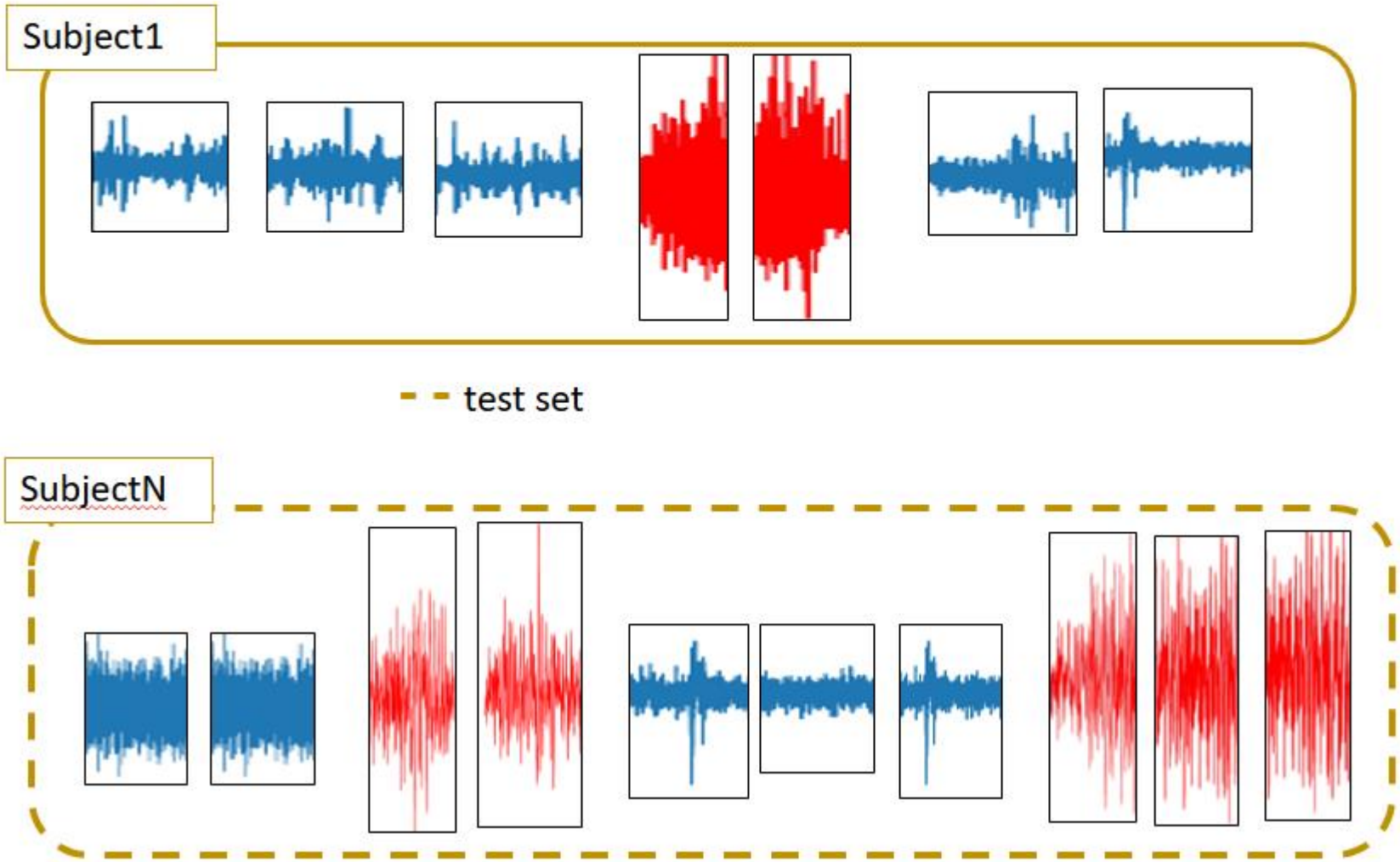
Patient Level

3. Data Split: Leave-one-patient out



Patient Level

3. Data Split: Leave-one-patient out



Getting started

1. Create GitHub repository for the group project. Share the link with me!
2. Assign roles: data scientist, data engineer, software engineer, project manager.
3. Access the data: */hhome/ricse/Epilepsy*
4. Read the data description. Material at virtual campus: *Description and Sample of EEG Recordings/DatasetDescription.pdf*
5. Start analysing the data!